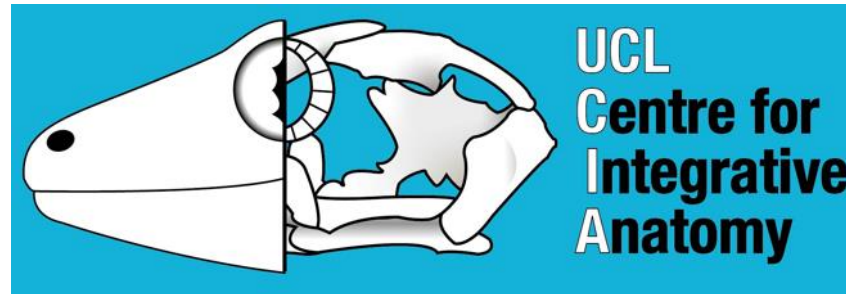
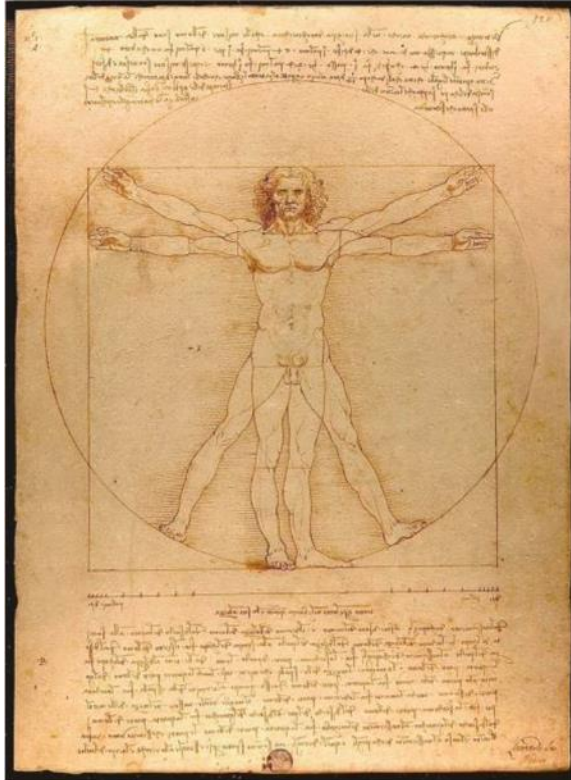


A Brief History of Geometric Morphometrics

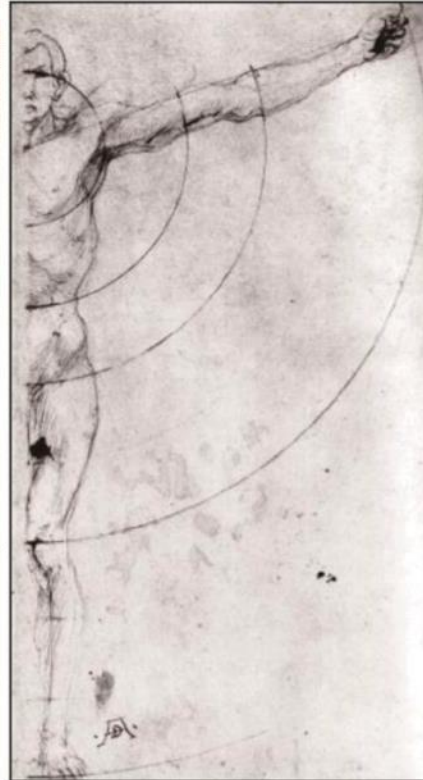
Ryan N. Felice



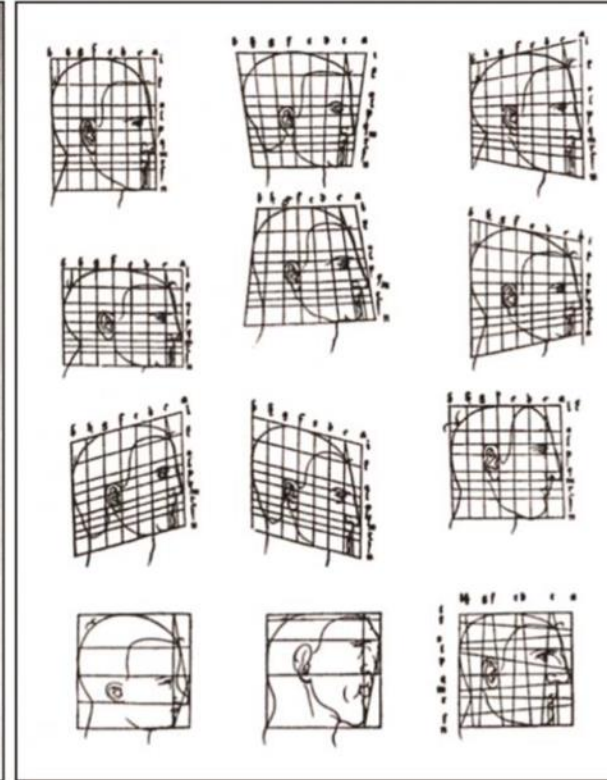
Analytical Paleobiology Workshop - Erlangen 2026



daVinci (c. 1480)

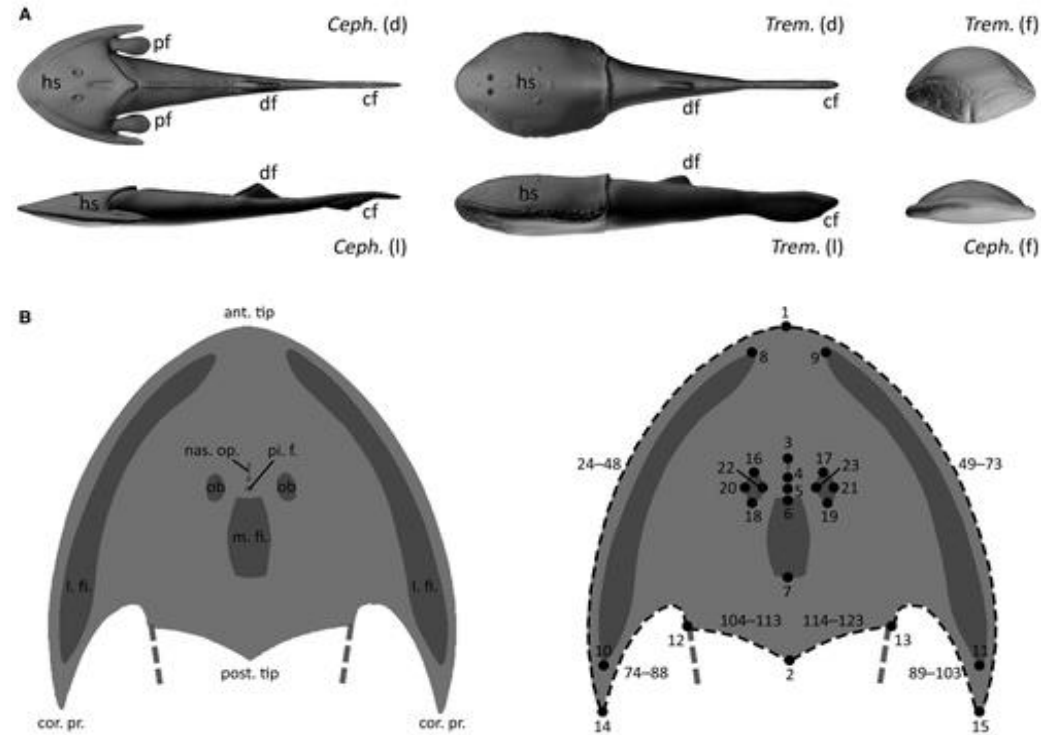


Düer (c. 1524)



Why Might Paleontologists Study Shape?

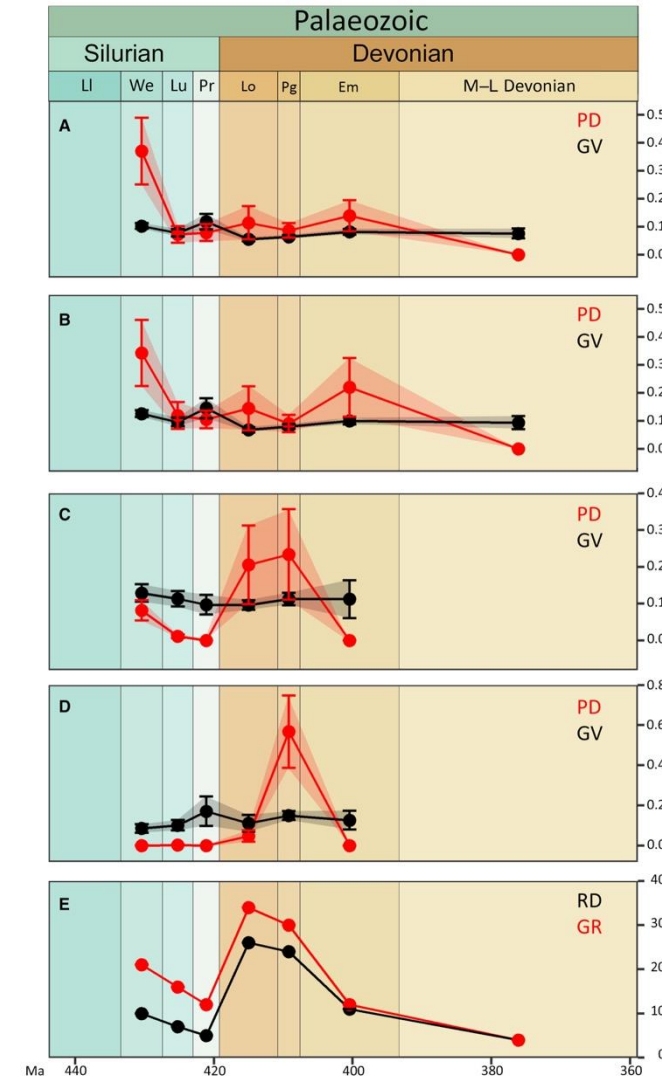
Disparity



Palaeontology The Palaeontological Association
www.palass.org
[Palaeontology, Vol. 63, Part 5, 2020, pp. 717–732]

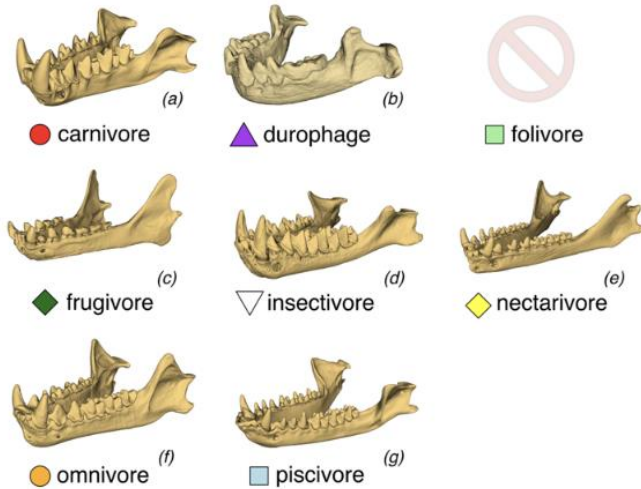
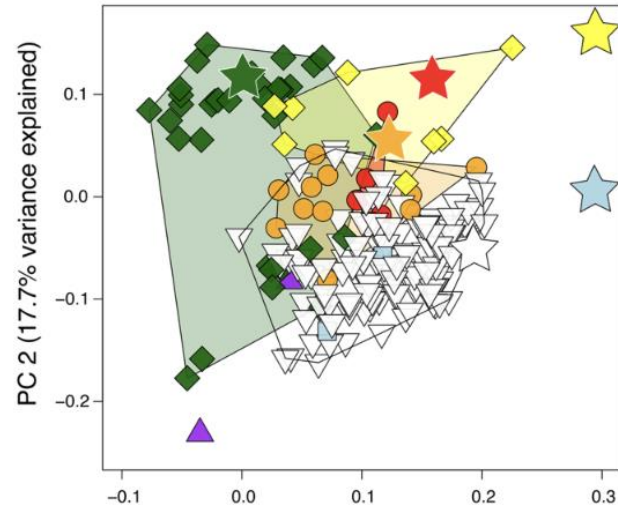
CATEGORICAL VERSUS GEOMETRIC
MORPHOMETRIC APPROACHES TO
CHARACTERIZING THE EVOLUTION OF
MORPHOLOGICAL DISPARITY IN OSTEOSTRACI
(VERTEBRATA, STEM GNATHOSTOMATA)

by HUMBERTO G. FERRÓN^{1,2*} , JENNY M. GREENWOOD¹,
BRADLEY DELINE³, CARLOS MARTÍNEZ-PÉREZ^{1,2}, HÉCTOR BOTELLA²,
ROBERT S. SANSOM⁴, MARCELLO RUTA⁵  and PHILIP C. J. DONOGHUE^{1,*} 

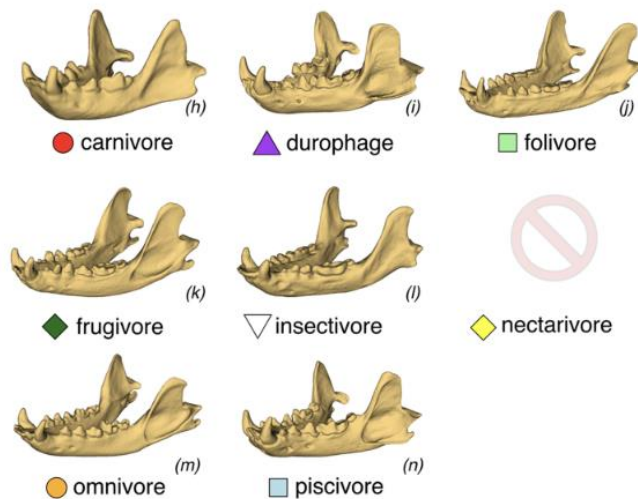
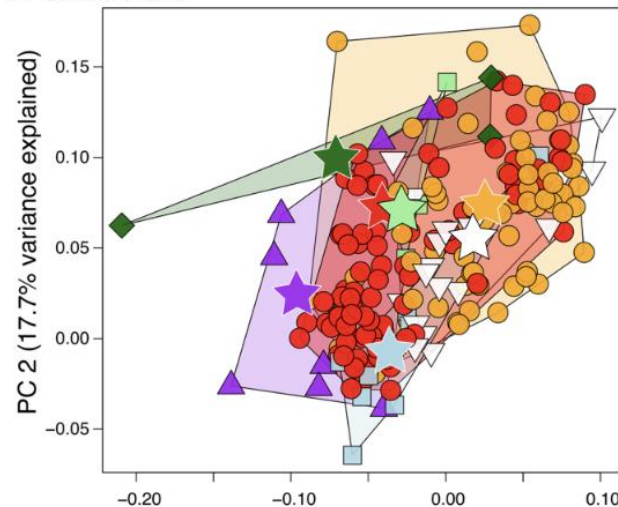


Ecomorphology

A. Bats



B. Carnivorans



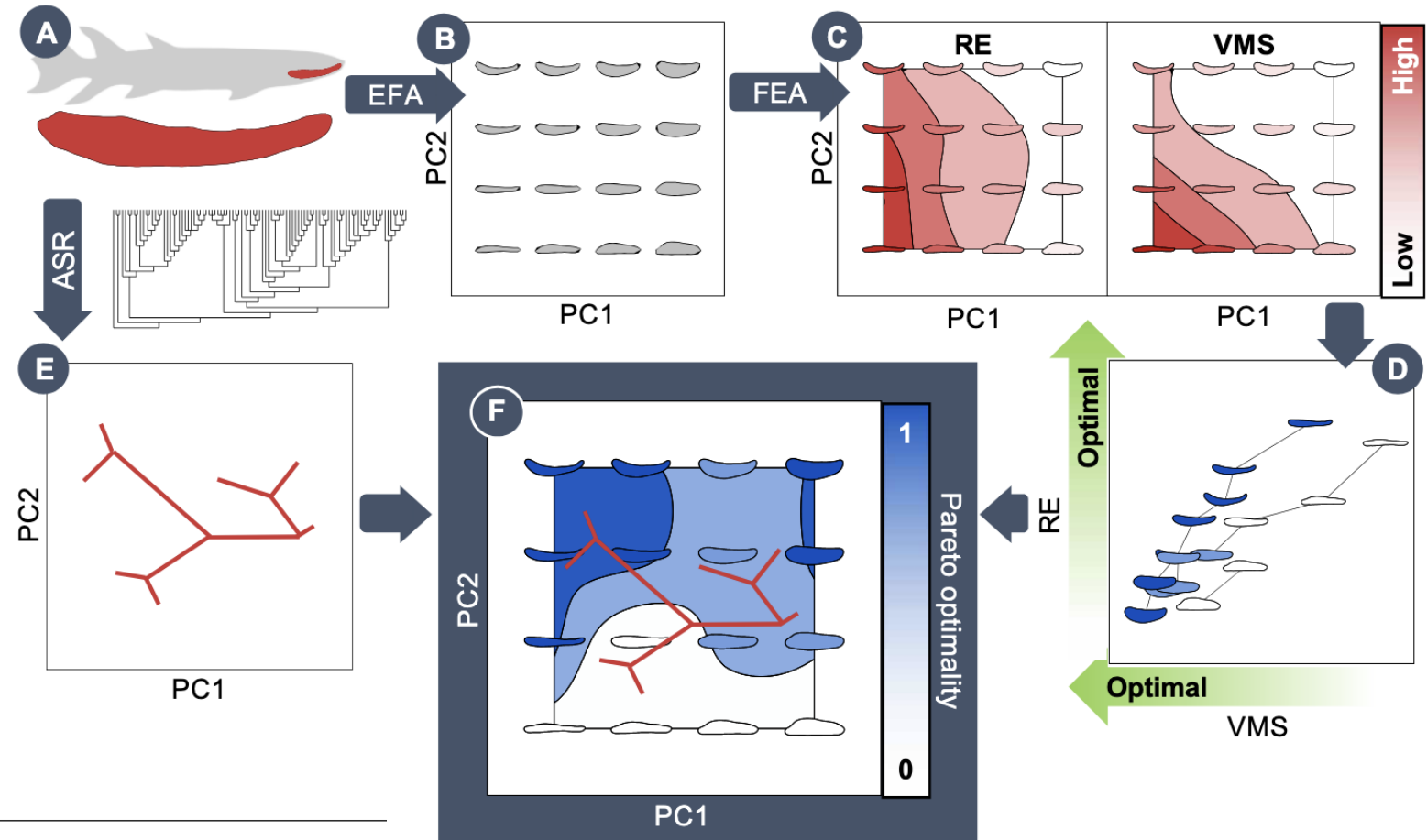
Article

Convergent function, divergent form: Shared diets do not produce shared adaptive zones in mammalian mandibular evolution

Chris Law, Michael Berthaume, Savannah Cobb, Sharlene Santana



Functional Morphology



SCIENCE ADVANCES | RESEARCH ARTICLE

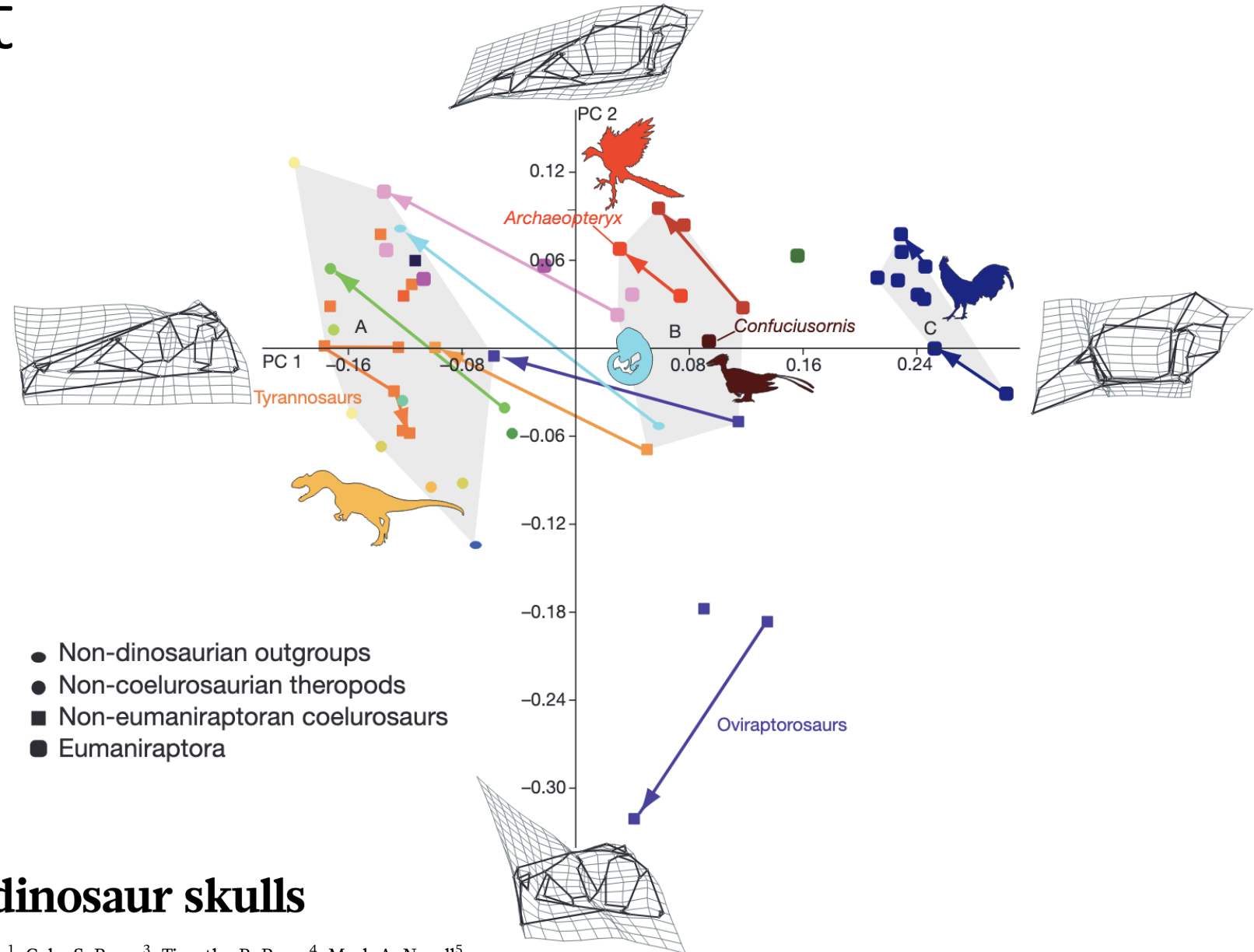
PALEONTOLOGY

Increasing morphological disparity and decreasing optimality for jaw speed and strength during the radiation of jawed vertebrates

William J. Deakin^{1*}, Philip S. L. Anderson², Wendy den Boer³, Thomas J. Smith¹, Jennifer J. Hill⁴, Martin Rücklin⁵, Philip C. J. Donoghue^{1*}, Emily J. Rayfield^{1*}

Macroevolution

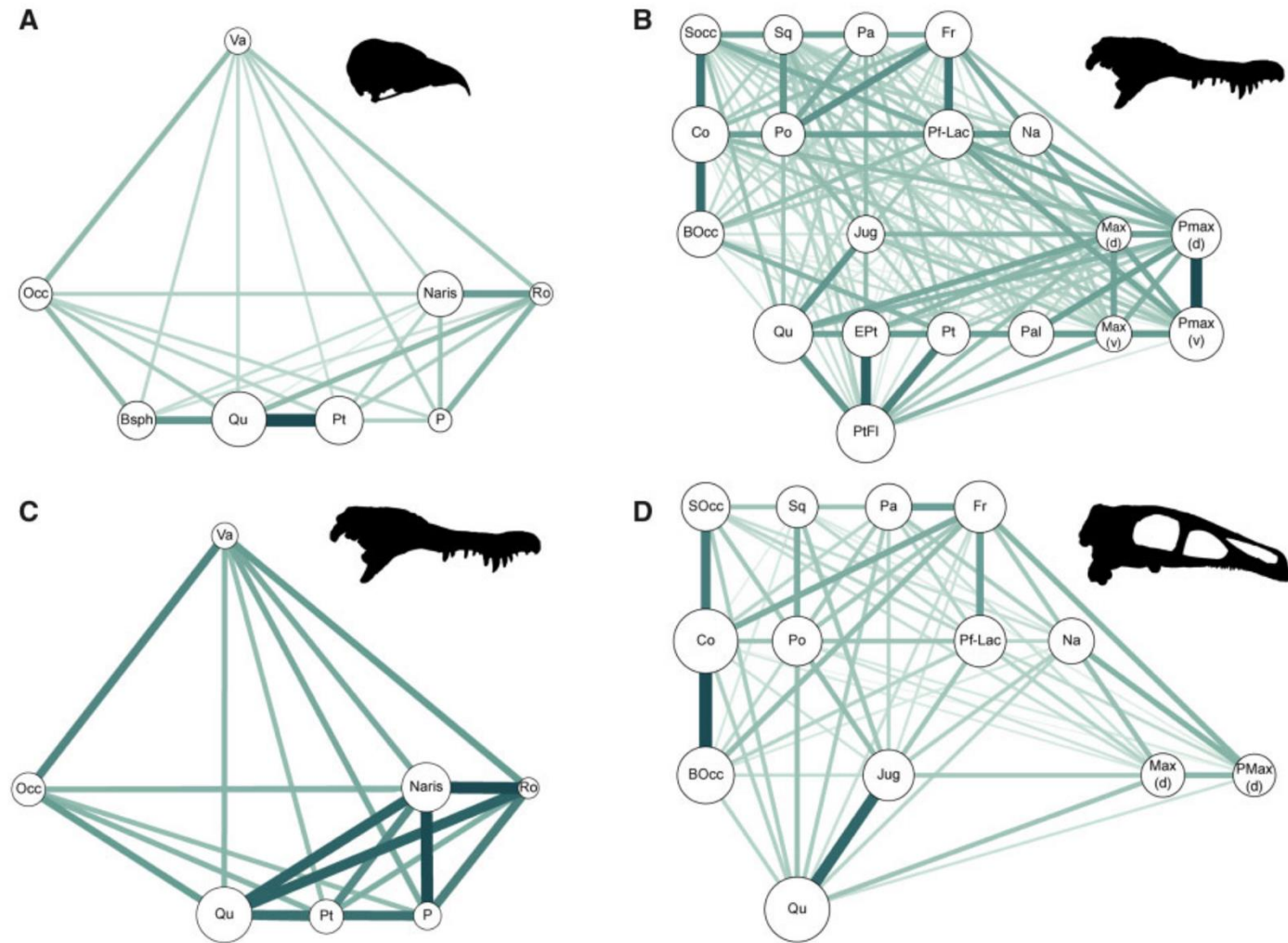
Development



Birds have paedomorphic dinosaur skulls

Bhart-Anjan S. Bhullar¹, Jesús Marugán-Lobón², Fernando Racimo¹, Gabe S. Bever³, Timothy B. Rowe⁴, Mark A. Norell⁵ & Arhat Abzhanov¹

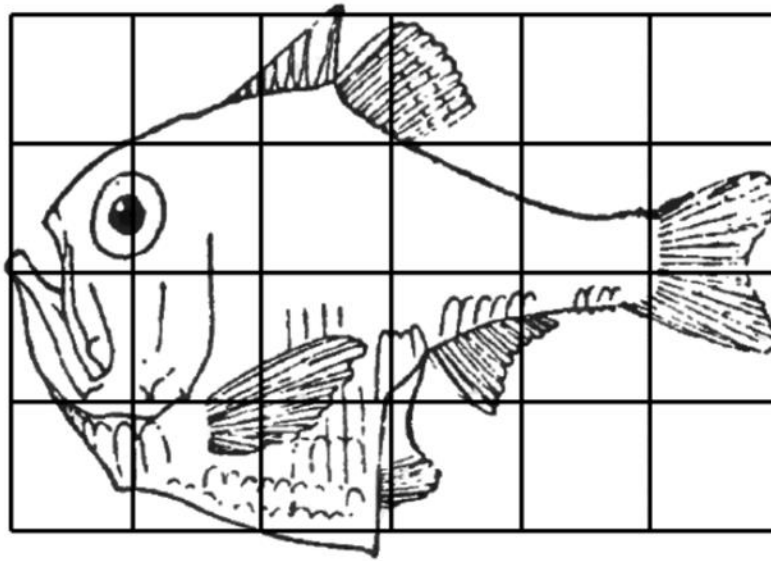
Modularity



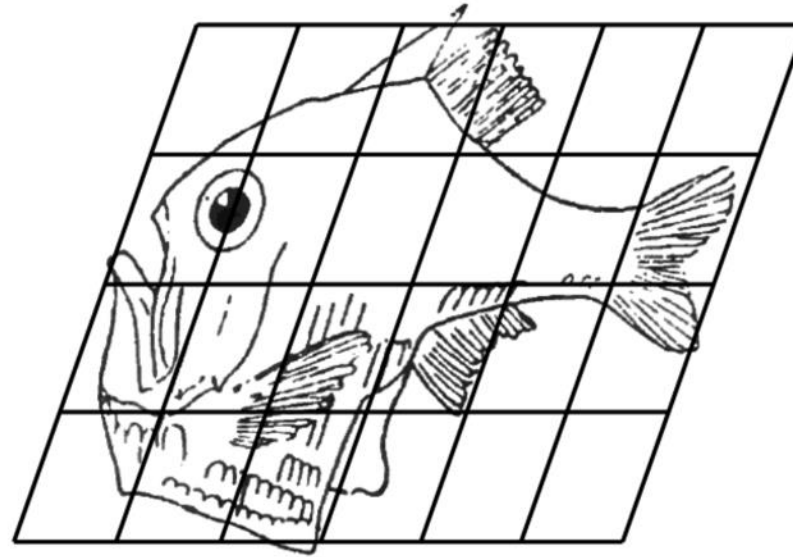
Evolutionary Integration and Modularity in the Archosaur Cranium

Ryan N. Felice,^{1,*†} Akinobu Watanabe,^{†,‡,§} Andrew R. Cuff,[¶] Eve Noirault,[†] Diego Pol,^{||}
 Lawrence M. Witmer,[#] Mark A. Norell,[§] Patrick M. O'Connor^{#,**} and Anjali Goswami^{†,††}

Shape Change

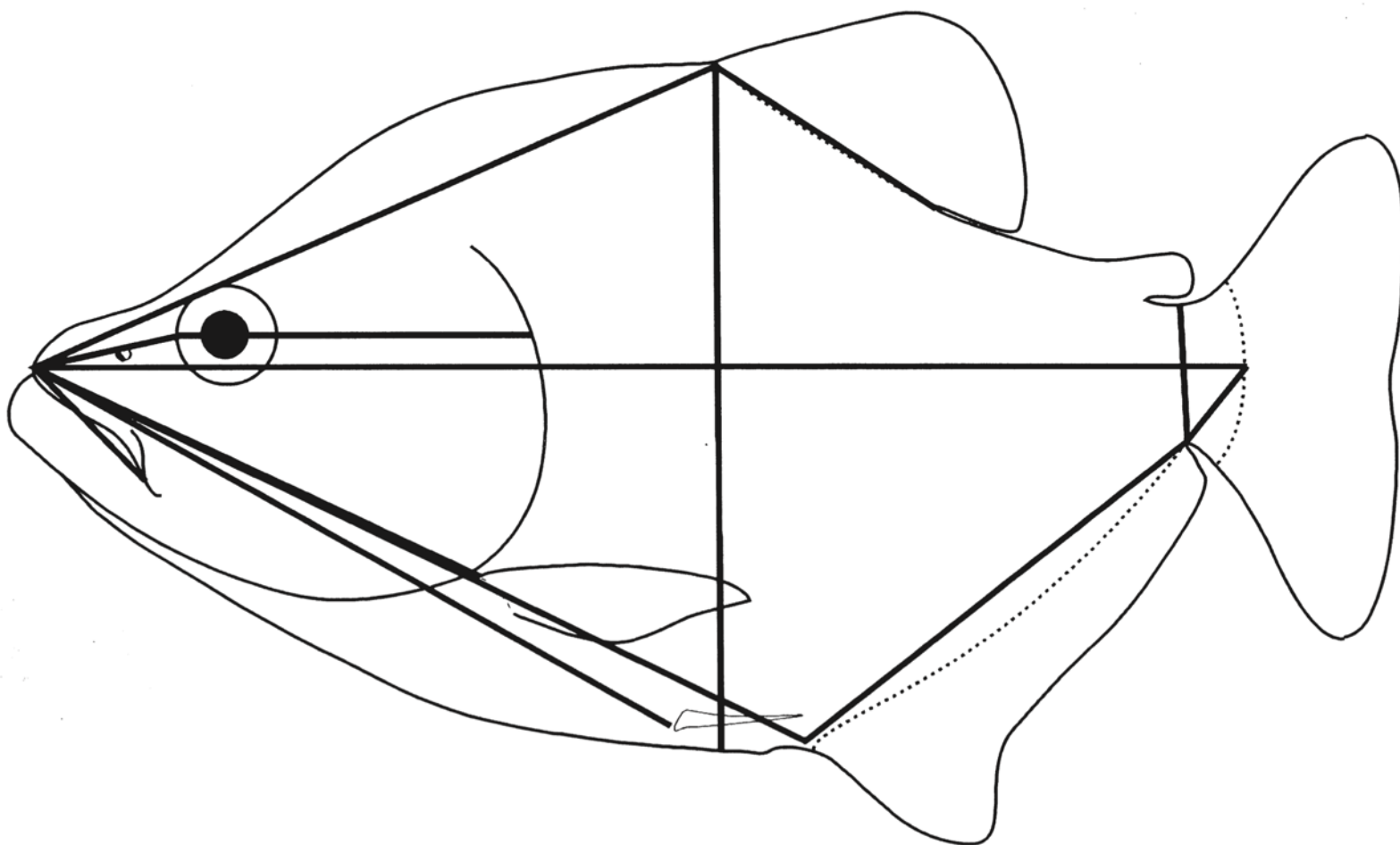


Argyroplecus



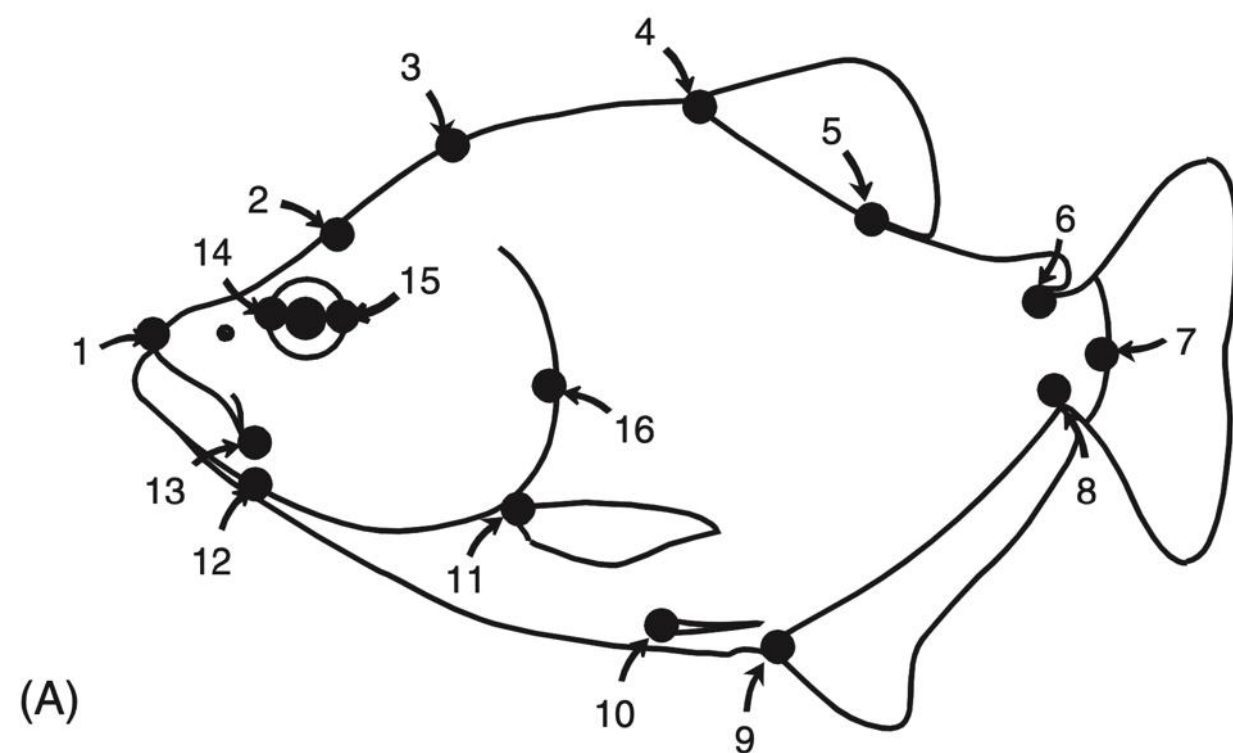
Sternopyx

D'Arcy Thompson

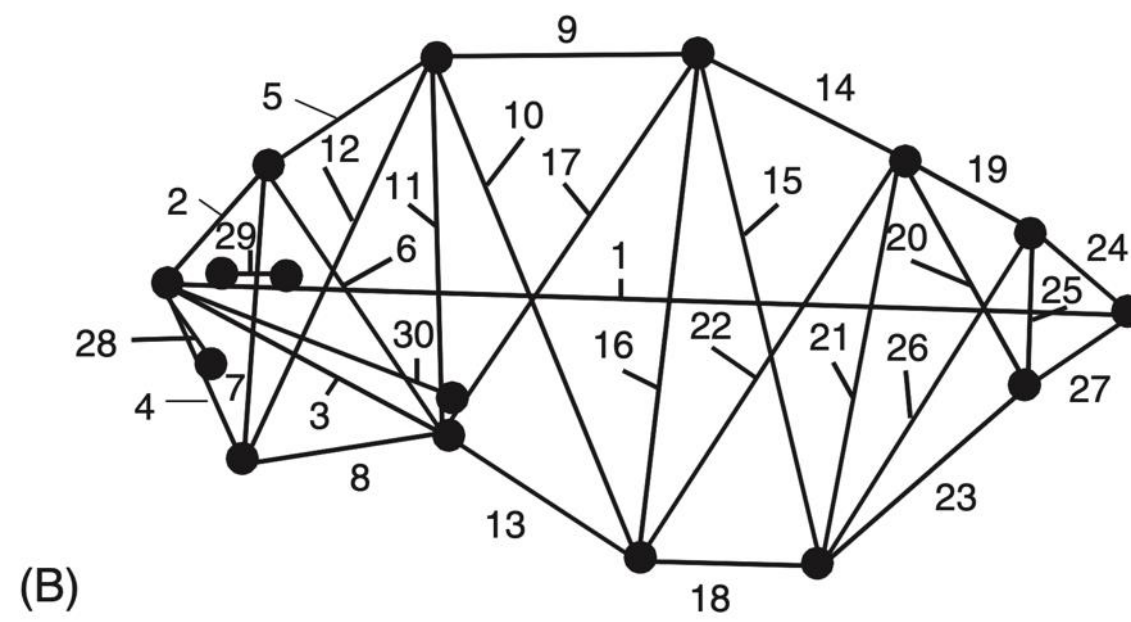


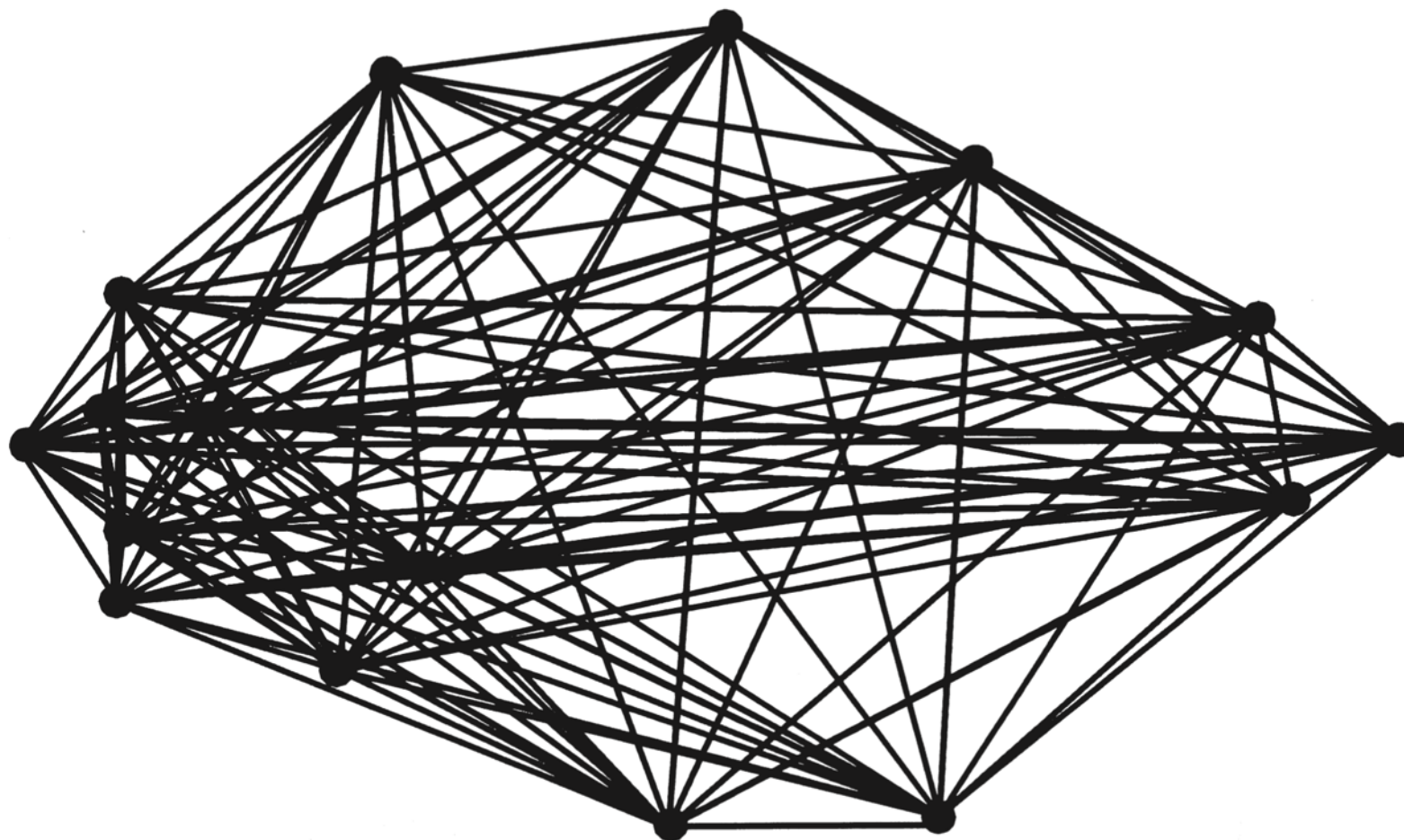
1930s/1940s

Truss measurements



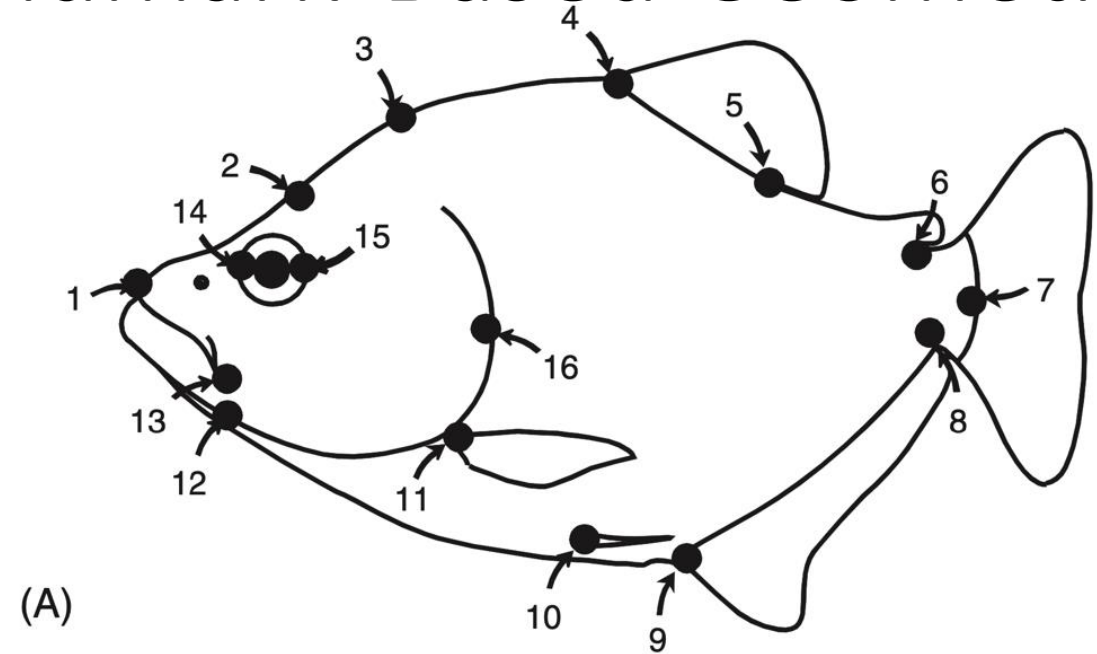
What is the drawback of this approach?



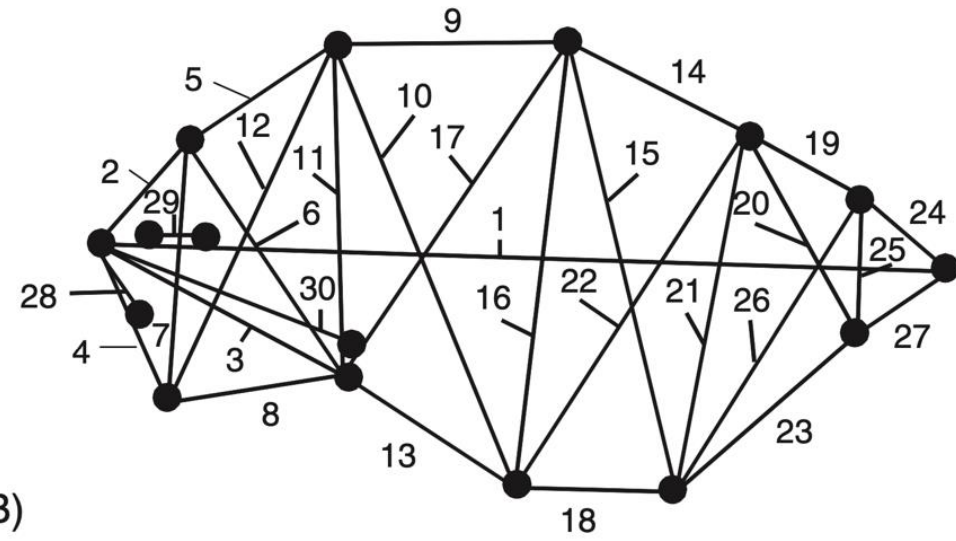


All pairwise truss measurements

The “Revolution”: Landmark-Based Geometric Morphometrics

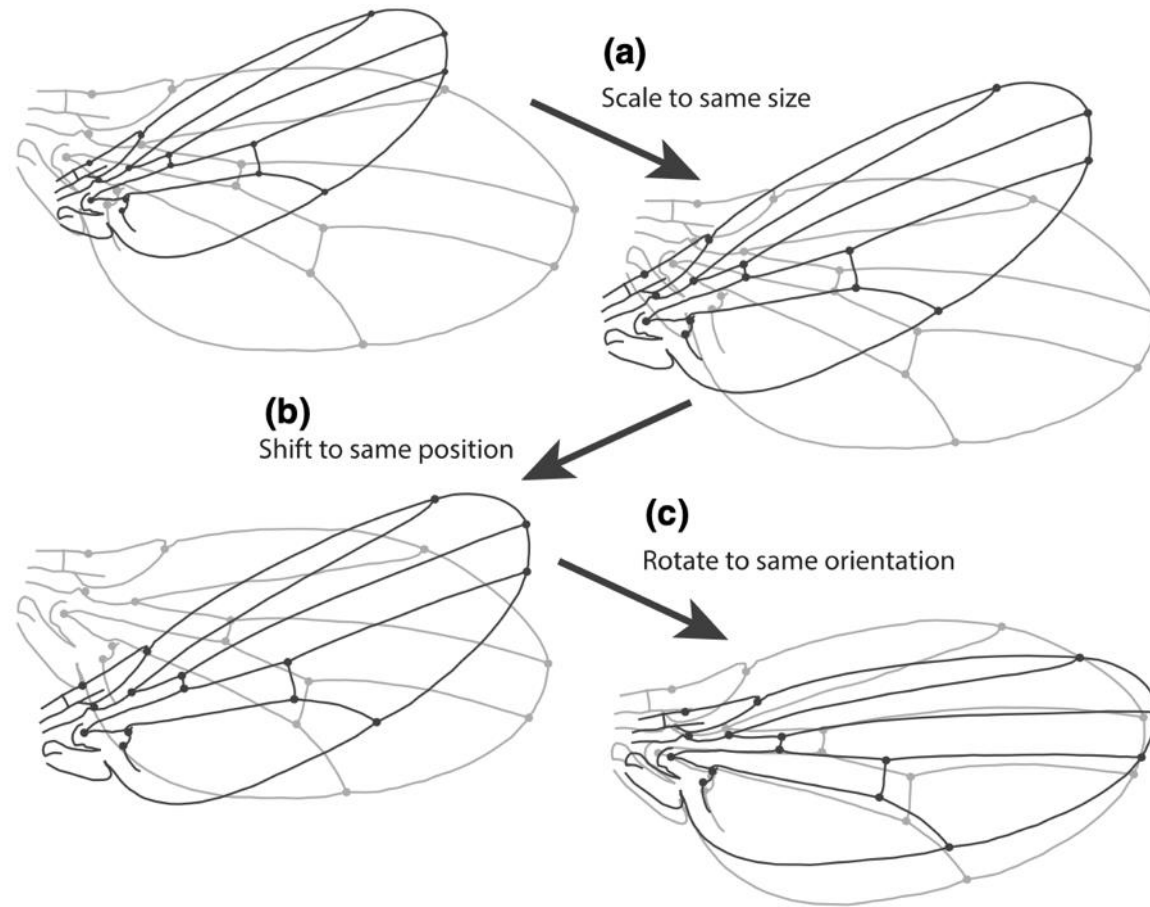


(A)



(B)

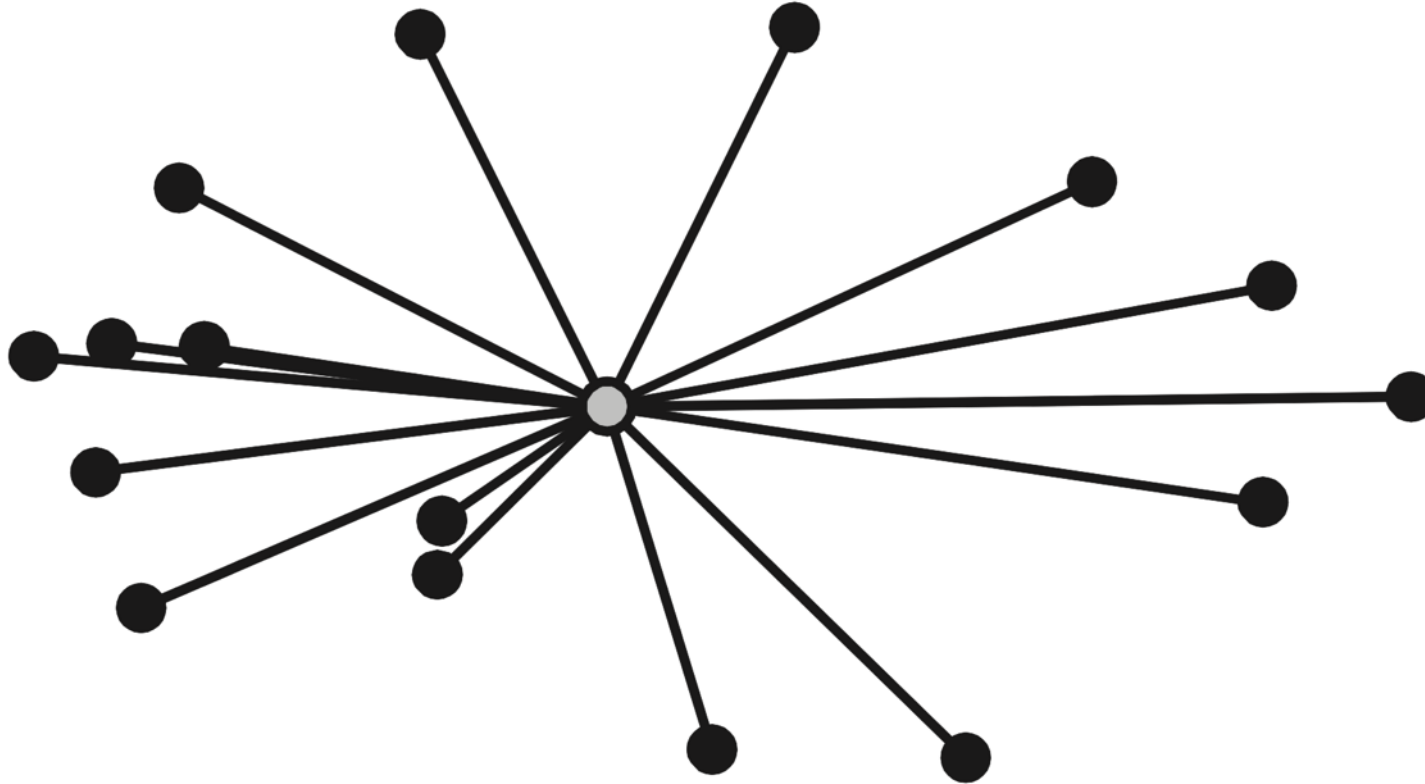
Procrustes superimposition



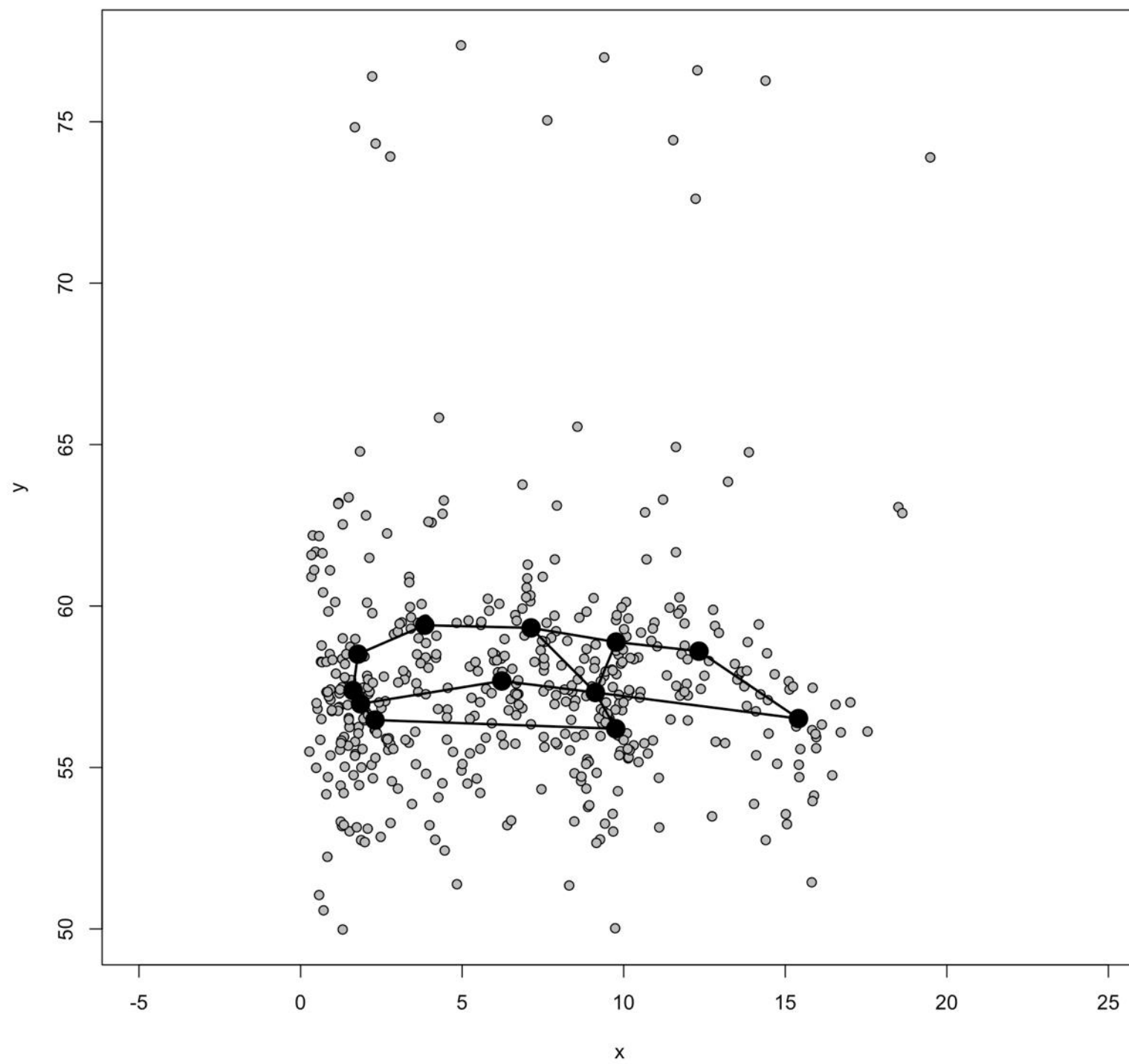


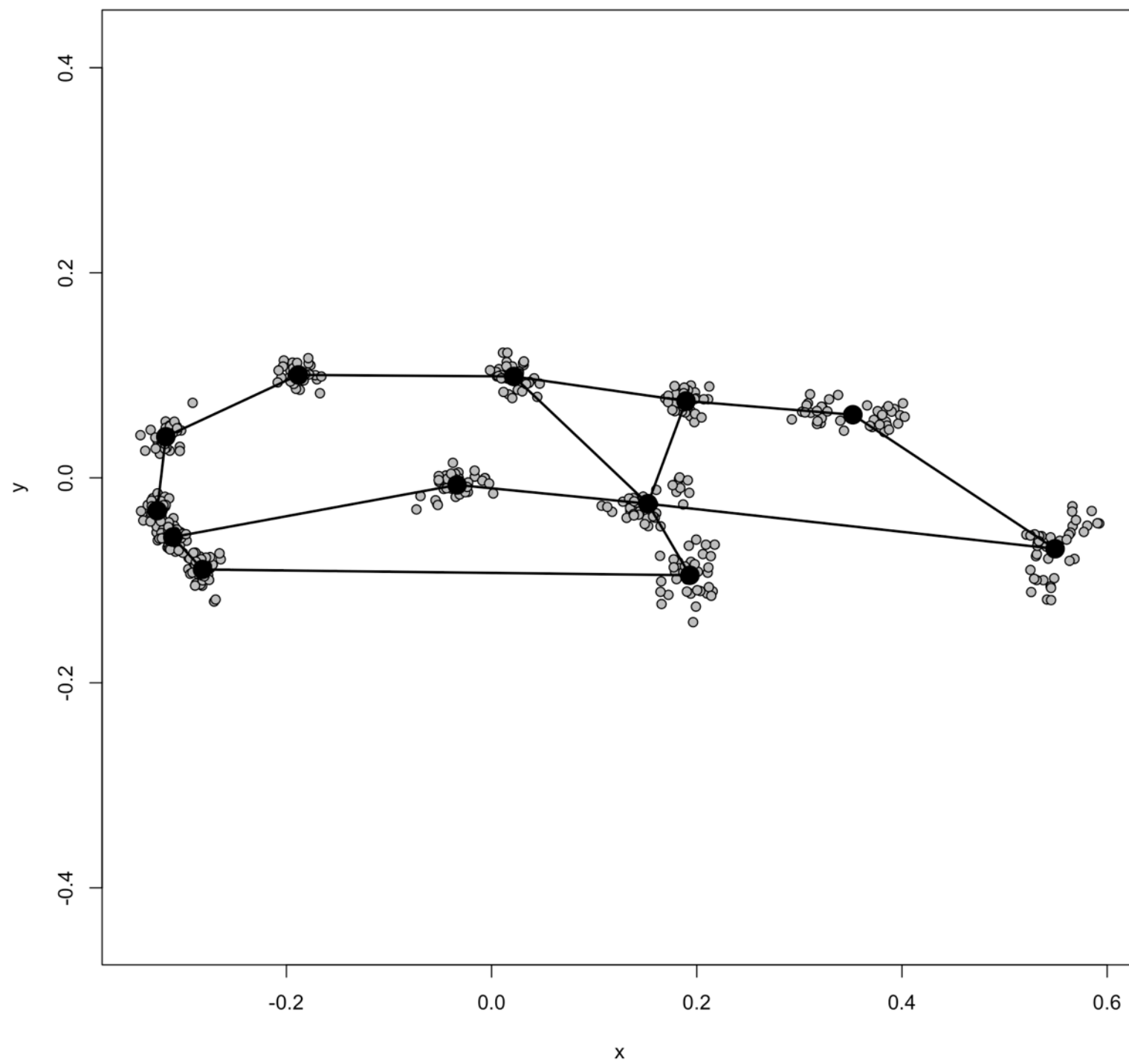
Procrustes had a stronghold on Mount Korydallos at Erineus, on the sacred way between Athens and Eleusis. There he had a bed, in which he invited every passer-by to spend the night, and where he set to work on them with his smith's hammer, to stretch them to fit. ... If the guest proved too tall, Procrustes would amputate the excess length; if the guest was too short Procrustes would stretch them until they died; nobody ever fit the bed exactly.

What is centroid size?



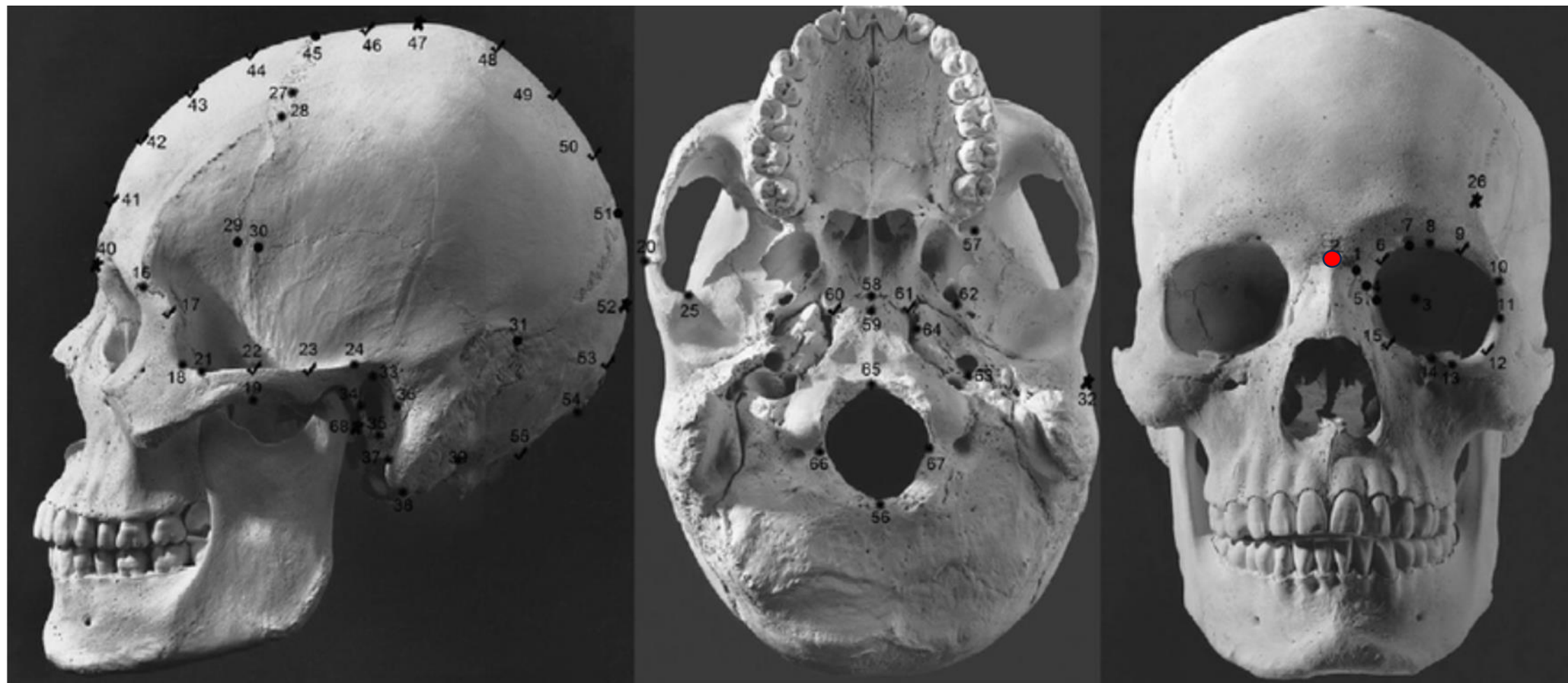
Sum of the squared distances from each point to the geometric center of all of the points

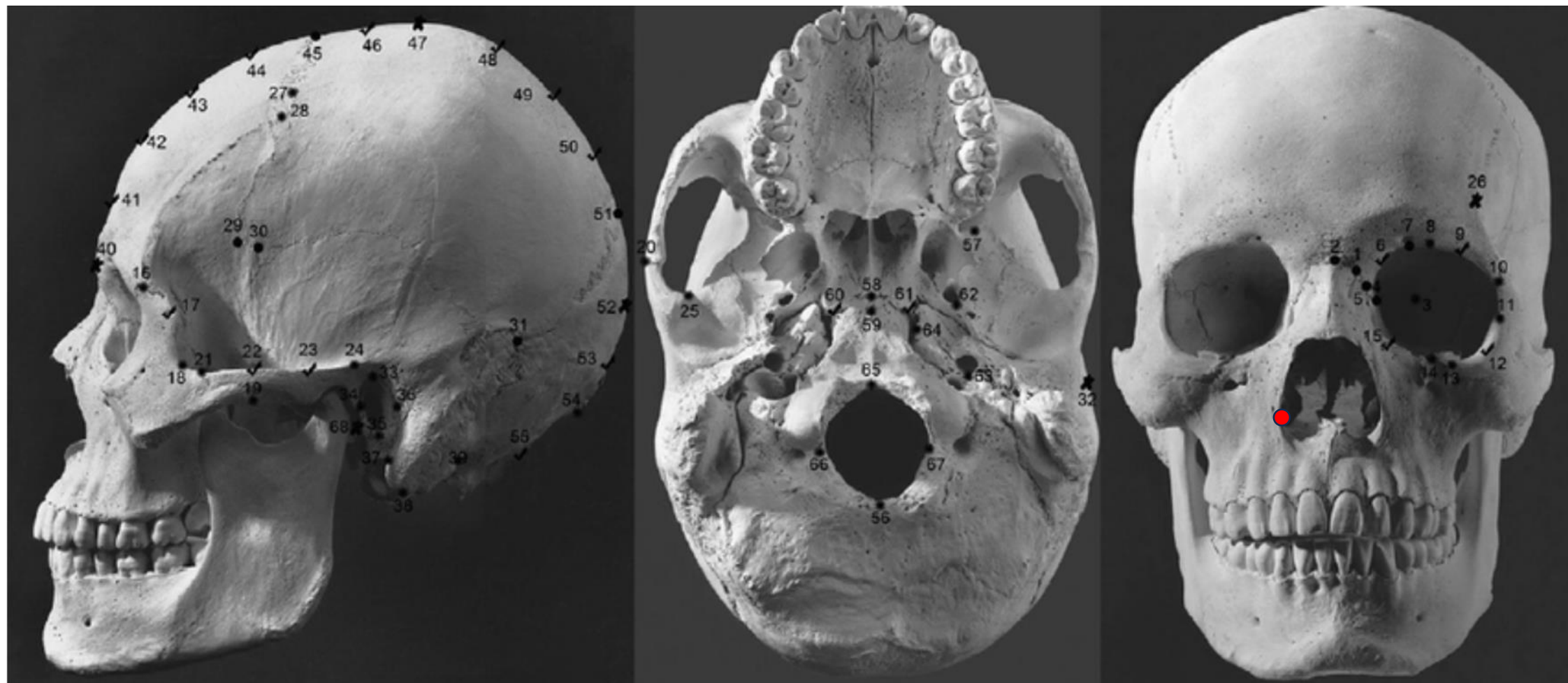


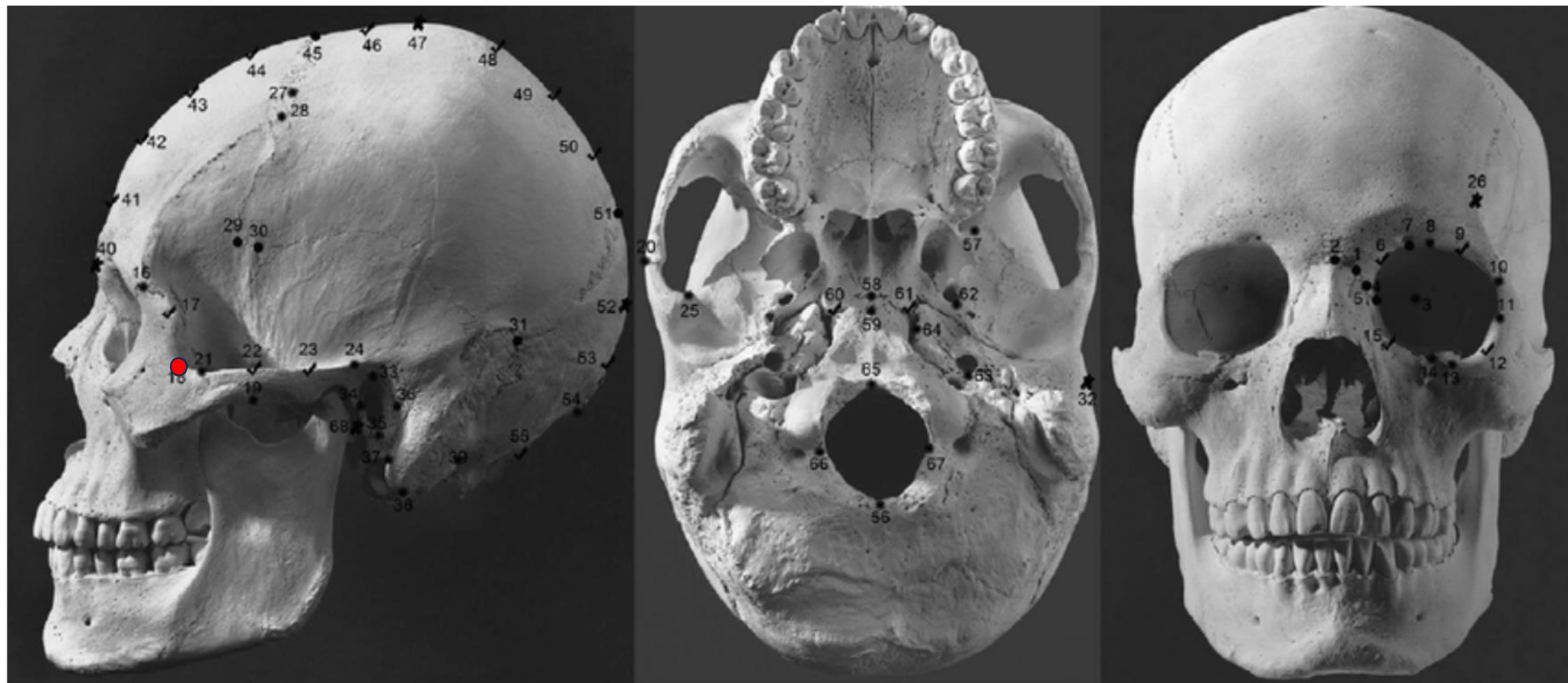


Landmark Types (Bookstein, 1991)

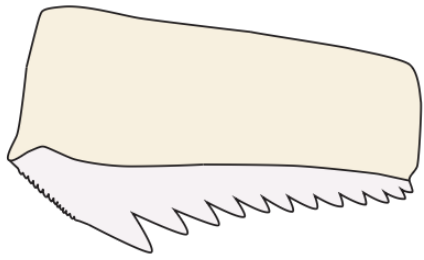
- **Type 1: Anatomical:** homology is provided by biologically unique patterns on the form (e.g. intersection of three bones)
- **Type 2: Processual:** homology is provided only by geometric, not biological or histological, criteria (e.g., point of maximum curvature along a boundary)
- **Type 3 extremal:** “end-points of diameters, centroids, intersections of interlandmark segments, points farthest from such segments, constructions involving perpendiculars or evenly spaced radial intercepts, and the like”



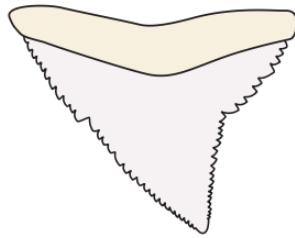




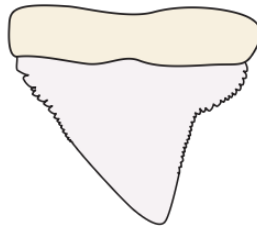
Alternatives to landmark-based morphometrics



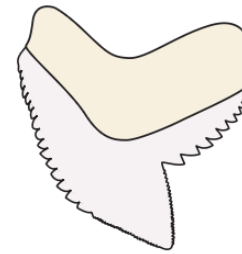
Hexanchus griseus



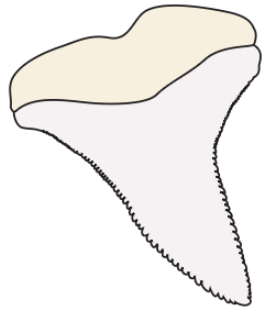
Carcharhinus falciformis



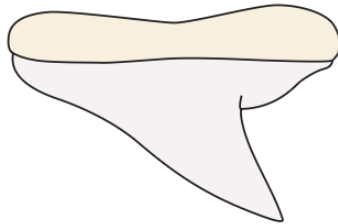
C. plumbeus



Galeocerdo cuvier



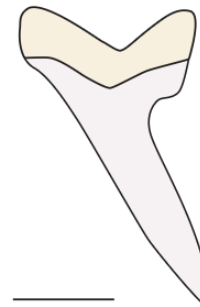
Prionace glauca



Sphyrna lewini



Ginglymostoma cirratum



Isurus oxyrinchus

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Research



Cite this article: Corn KA, Farina SC, Brash J, Summers AP. 2016 Modelling tooth–prey interactions in sharks: the importance of dynamic testing. *R. Soc. open sci.* 3: 160141. <http://dx.doi.org/10.1098/rsos.160141>

Modelling tooth–prey interactions in sharks: the importance of dynamic testing

Katherine A. Corn¹, Stacy C. Farina², Jeffrey Brash³ and Adam P. Summers⁴

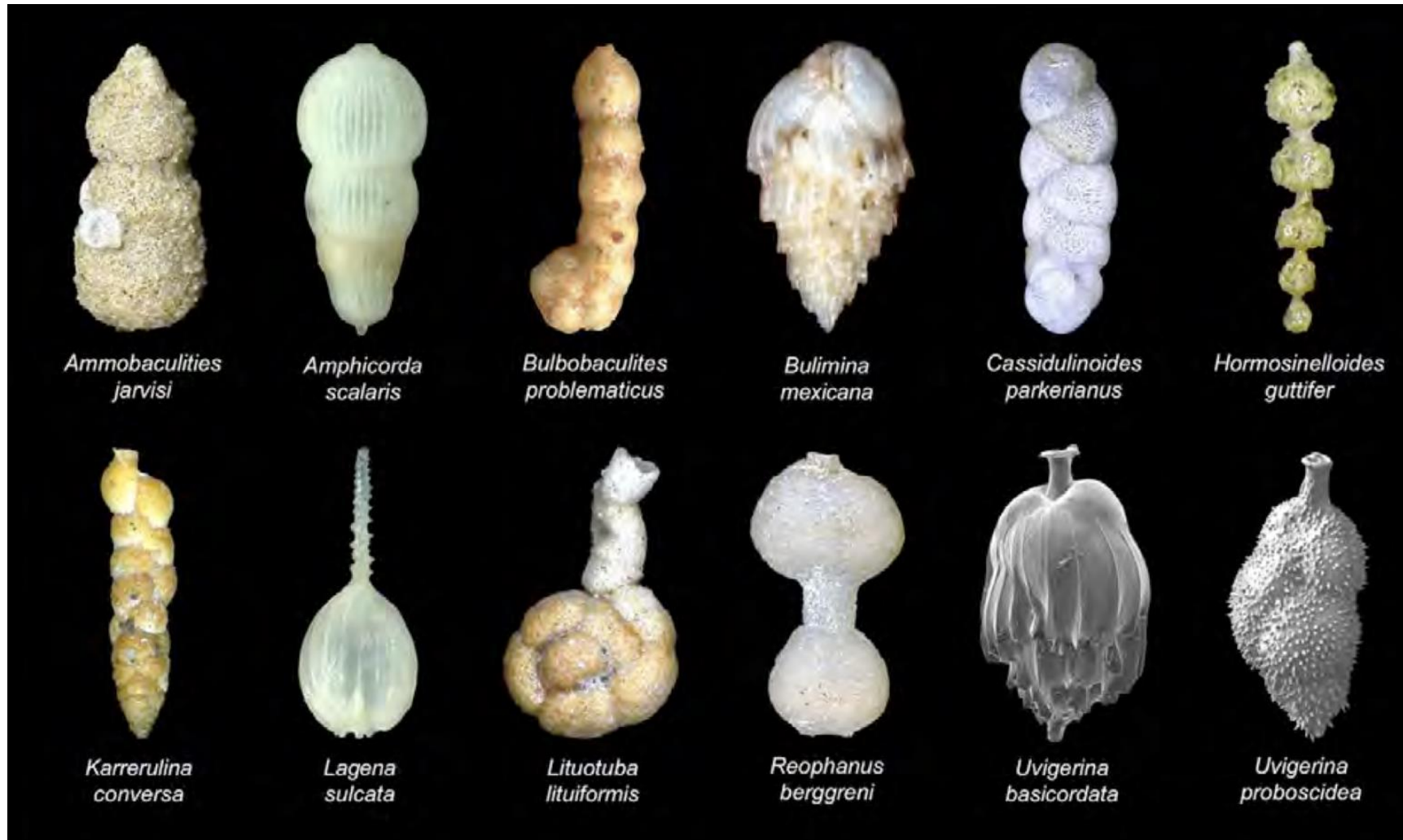
¹Department of Ecology and Evolutionary Biology, Cornell University, Ithaca, NY, USA

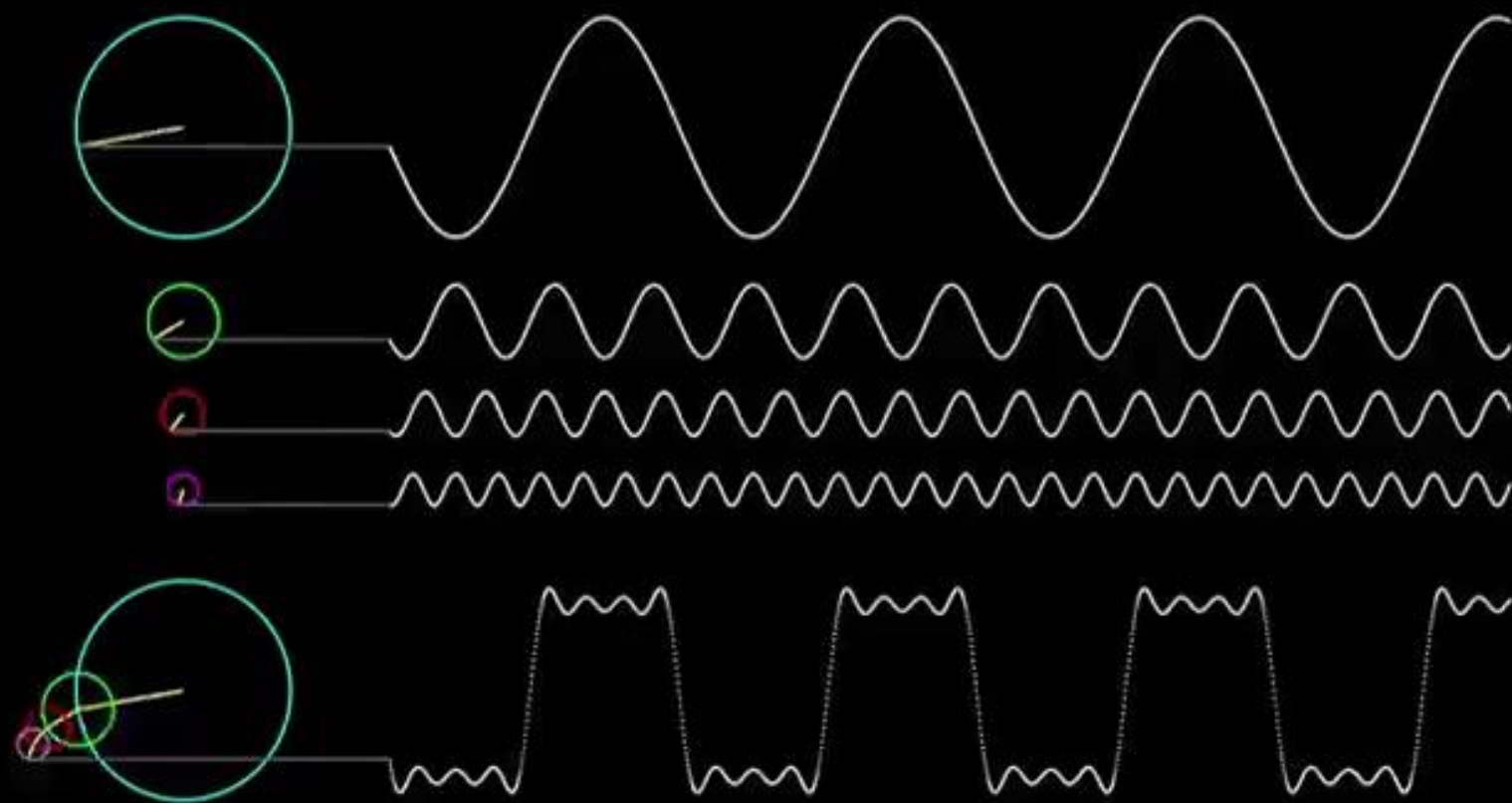
²Museum of Comparative Zoology, Harvard University, Cambridge, MA, USA

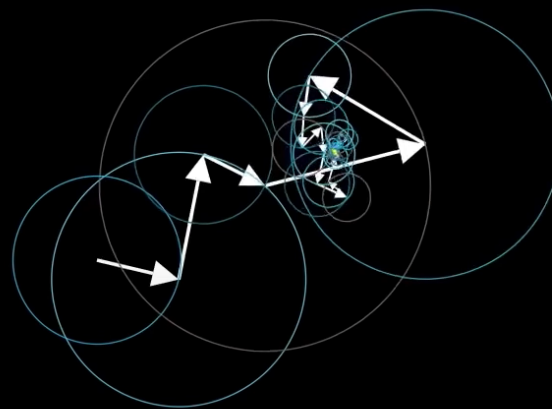
³Valley Steel and Stone, Friday Harbor, WA, USA

⁴School of Aquatic and Fisheries Sciences, University of Washington, Friday Harbor, WA, USA

Alternatives to landmark-based morphometrics



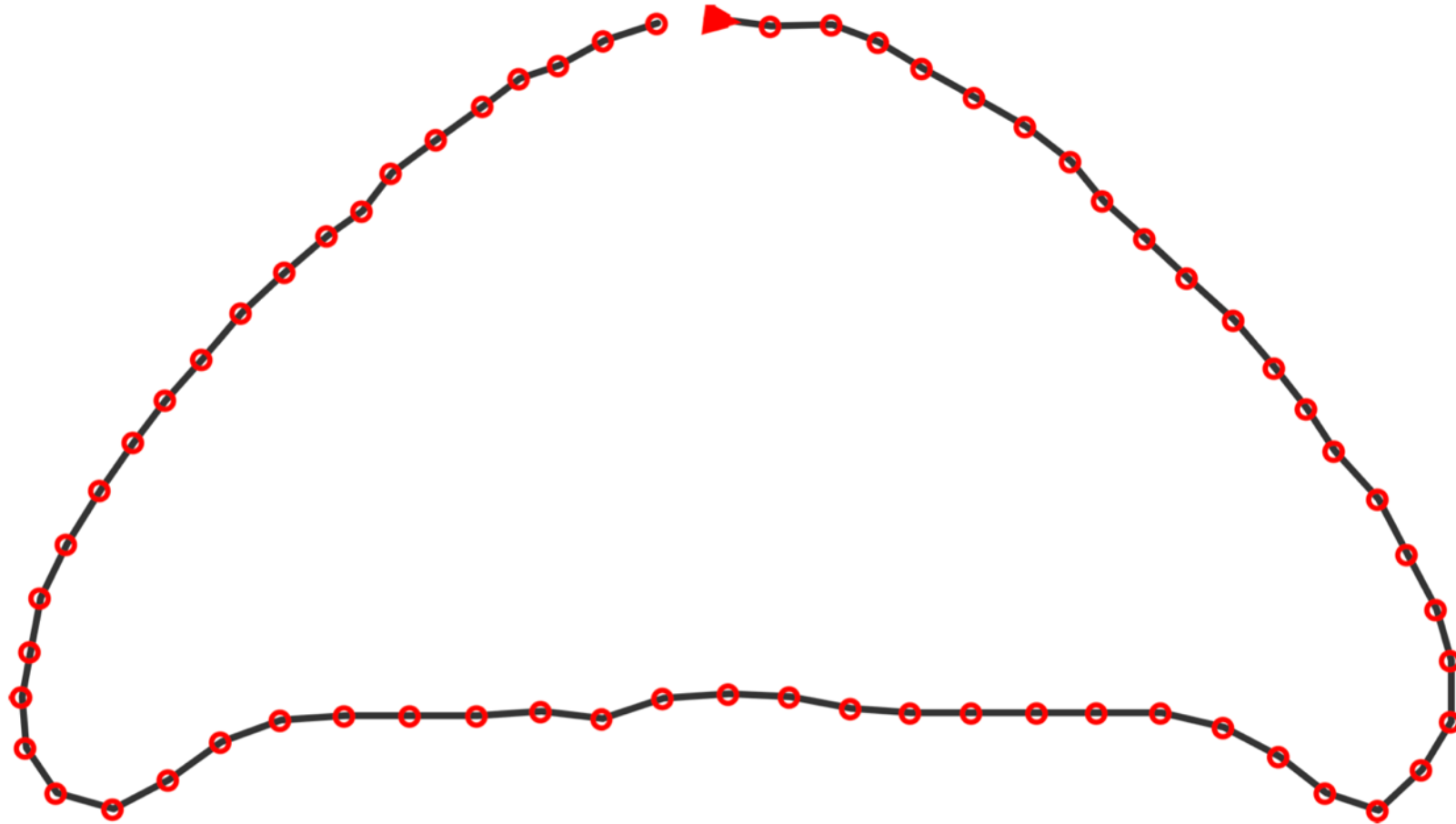


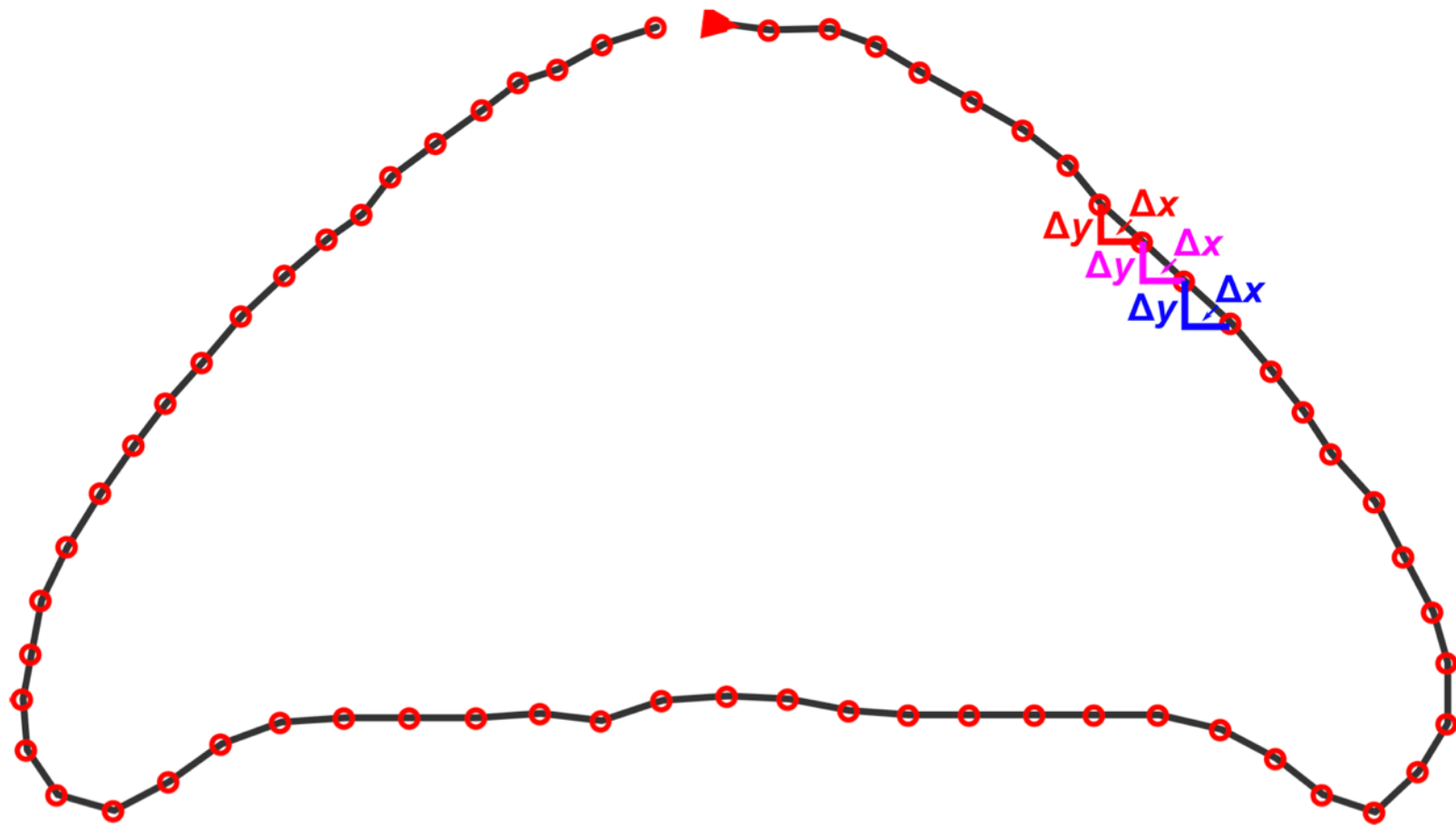


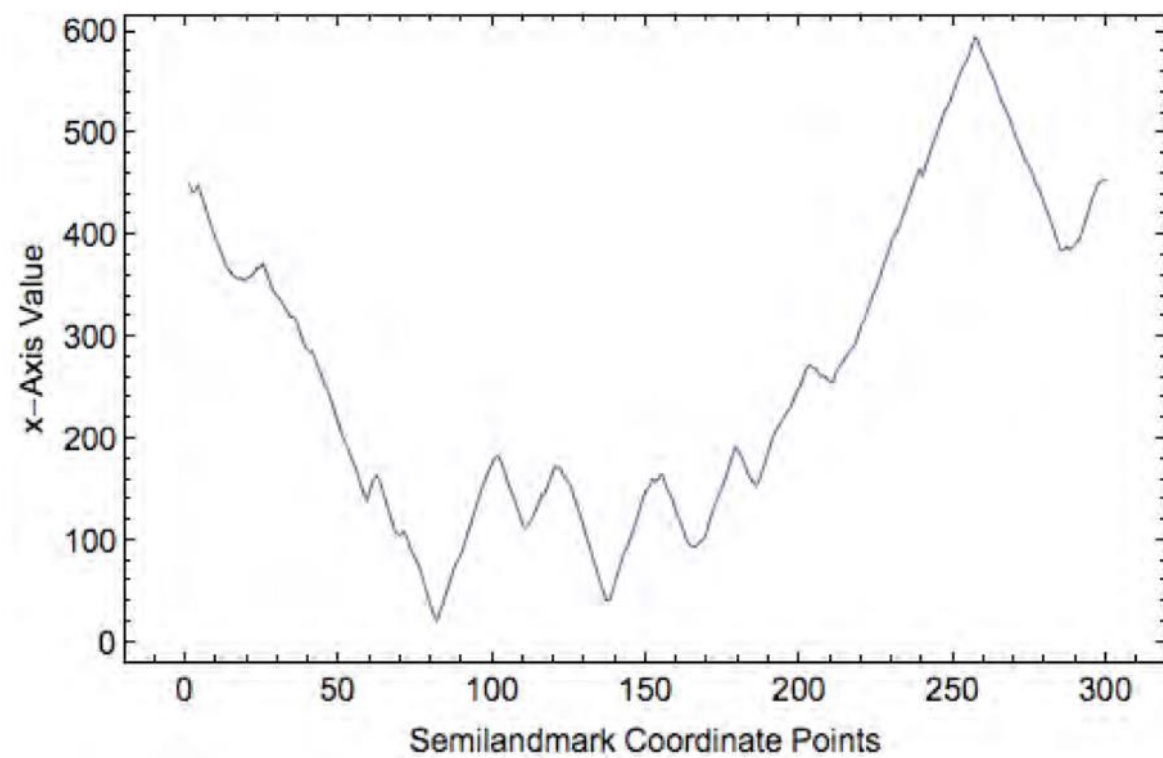
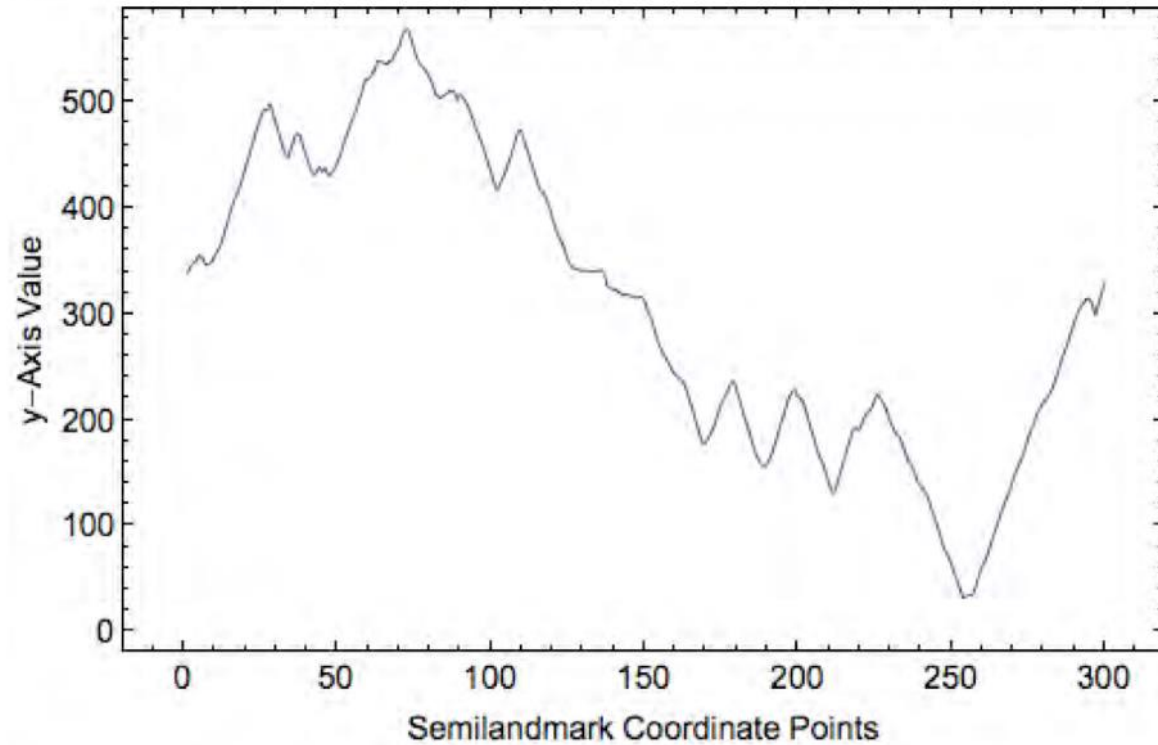
Play (k)



1960s/1970s- Outline analysis





A.**B.**

$$A_n = \frac{T}{2n^2\pi^2} \sum_{p=1}^k \frac{\Delta x_p}{\Delta t_p} \left[\cos\left(\frac{2\pi n t_p}{T}\right) - \cos\left(\frac{2\pi n t_{p-1}}{T}\right) \right]$$

$$B_n = \frac{T}{2n^2\pi^2} \sum_{p=1}^k \frac{\Delta x_p}{\Delta t_p} \left[\sin\left(\frac{2\pi n t_p}{T}\right) - \sin\left(\frac{2\pi n t_{p-1}}{T}\right) \right]$$

$$C_n = \frac{T}{2n^2\pi^2} \sum_{p=1}^k \frac{\Delta y_p}{\Delta t_p} \left[\cos\left(\frac{2\pi n t_p}{T}\right) - \cos\left(\frac{2\pi n t_{p-1}}{T}\right) \right]$$

$$D_n = \frac{T}{2n^2\pi^2} \sum_{p=1}^k \frac{\Delta y_p}{\Delta t_p} \left[\sin\left(\frac{2\pi n t_p}{T}\right) - \sin\left(\frac{2\pi n t_{p-1}}{T}\right) \right]$$

Where: k = the total number of steps around the outline

n = the harmonic number

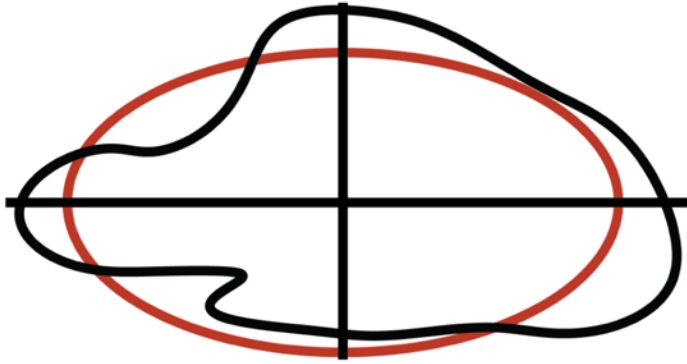
Δx = the displacement along the x axis between
point p and $p+1$

Δt = the length of the step between point p and $p+1$

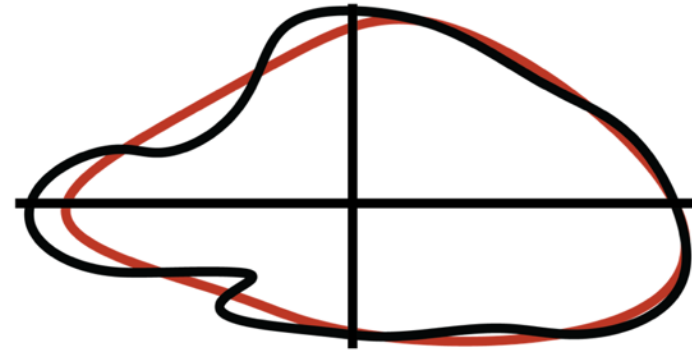
t_p = accumulated length of step segments at point p

T = sum of lengths of all steps around outline

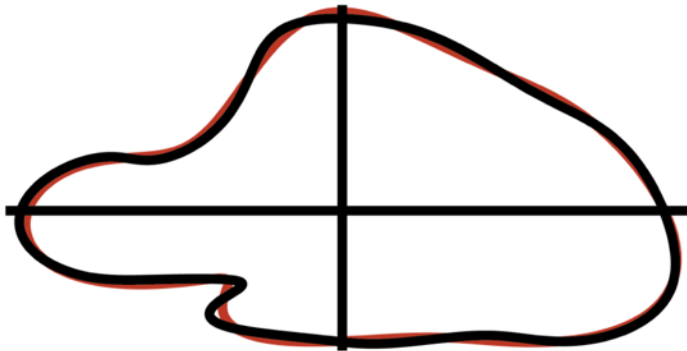
One Harmonic



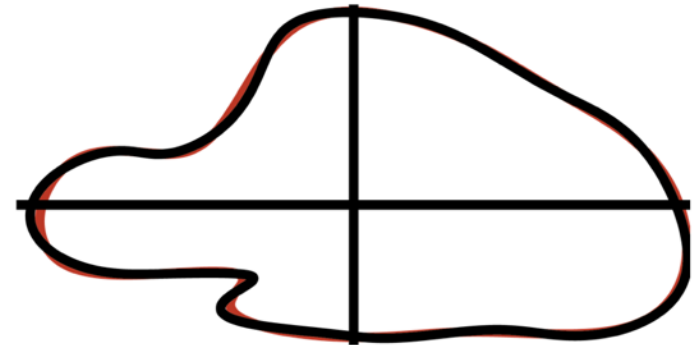
Three Harmonics



Six Harmonics



Eight Harmonics



3D version

Heart Chambers

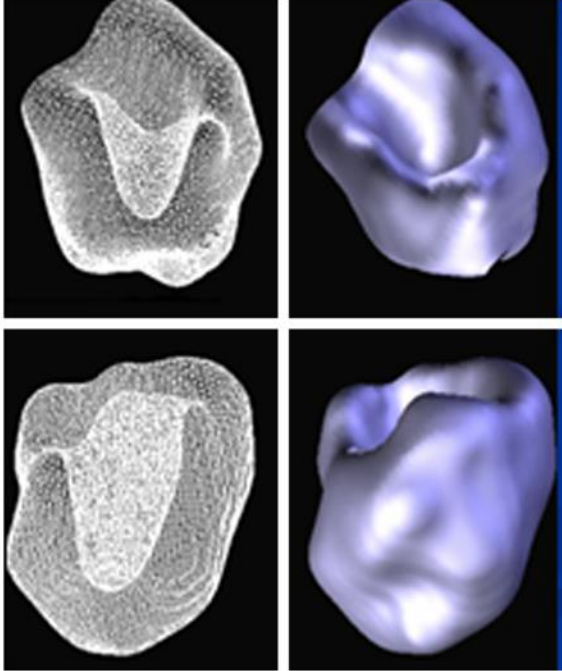


Fig. 19. Segmentation and separation of scintigraphic image sequences in one patient resulting in the structures of the epicardium wall and the endocardium wall.

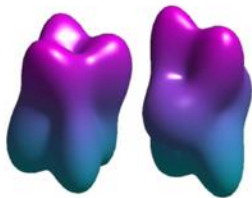


Fig. 20. Rendered surface of the left ventricle at stress and rest respectively after segmentation of scintigraphic image sequences in one patient.

Ben Abdallah et al 2010

Hippocampus

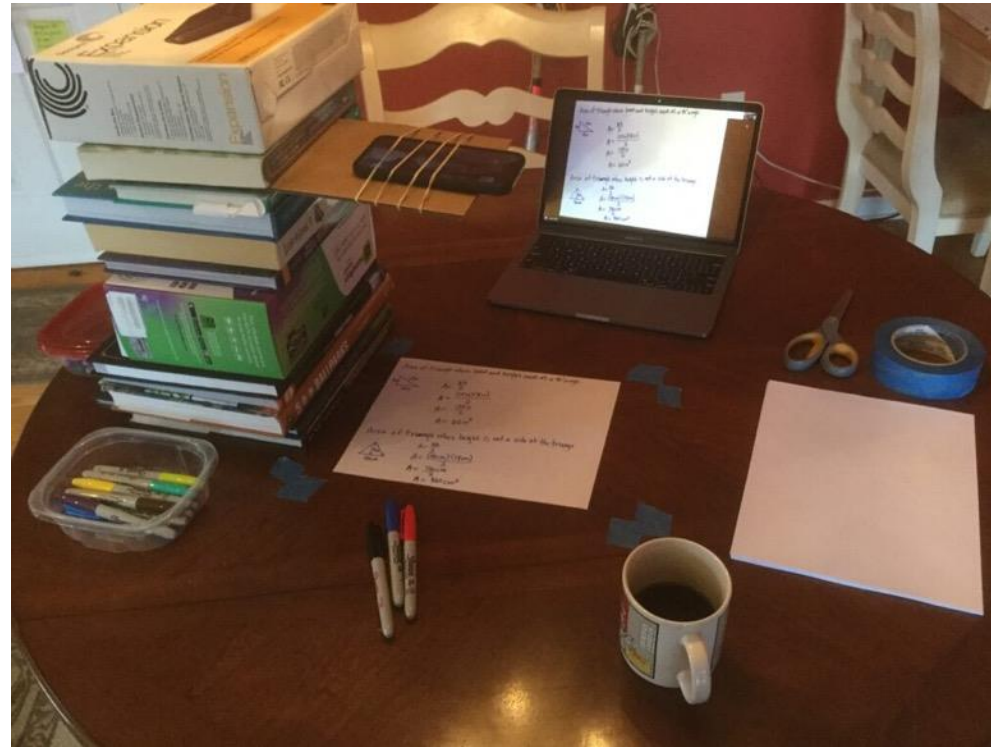


Gutman et al 2006

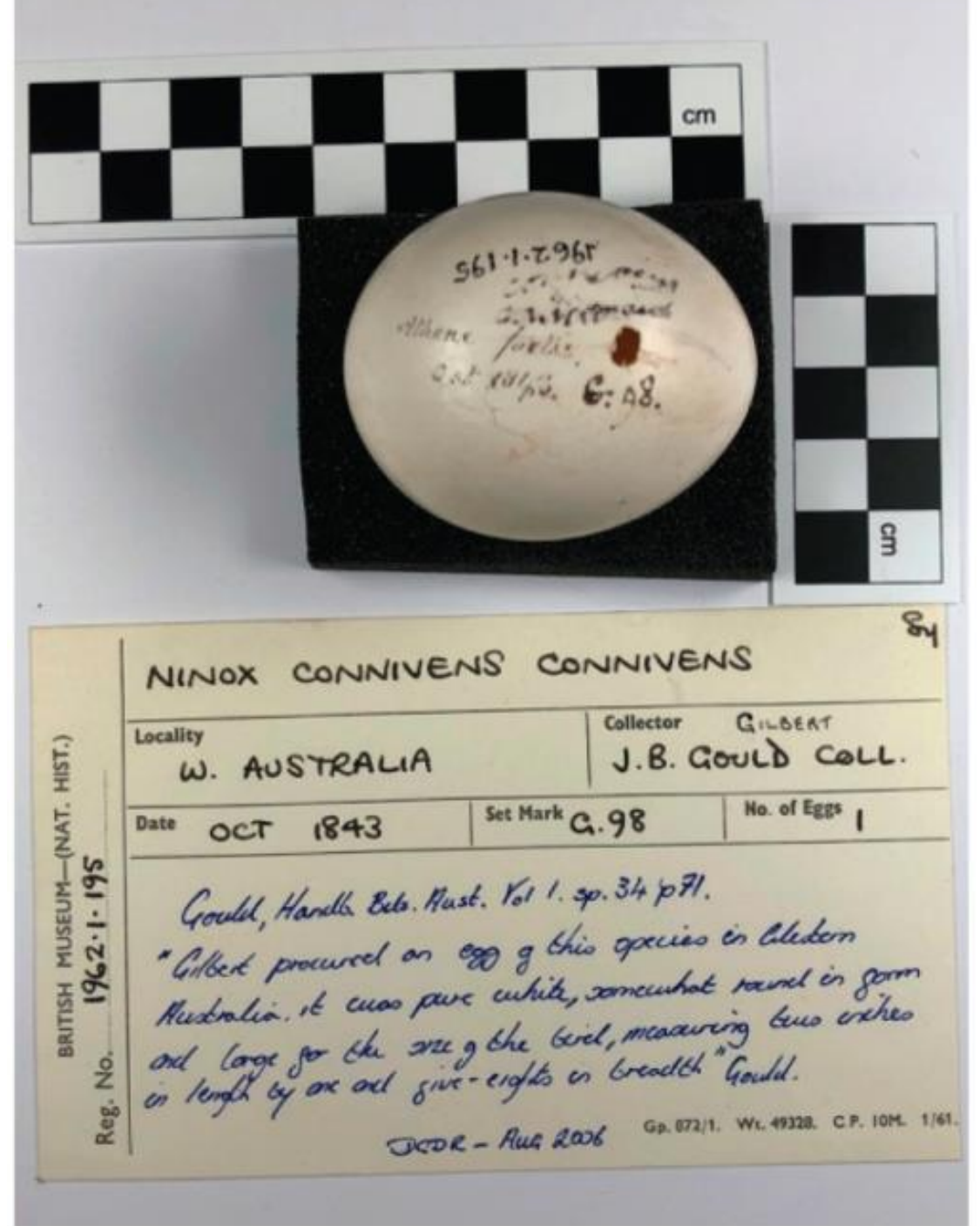
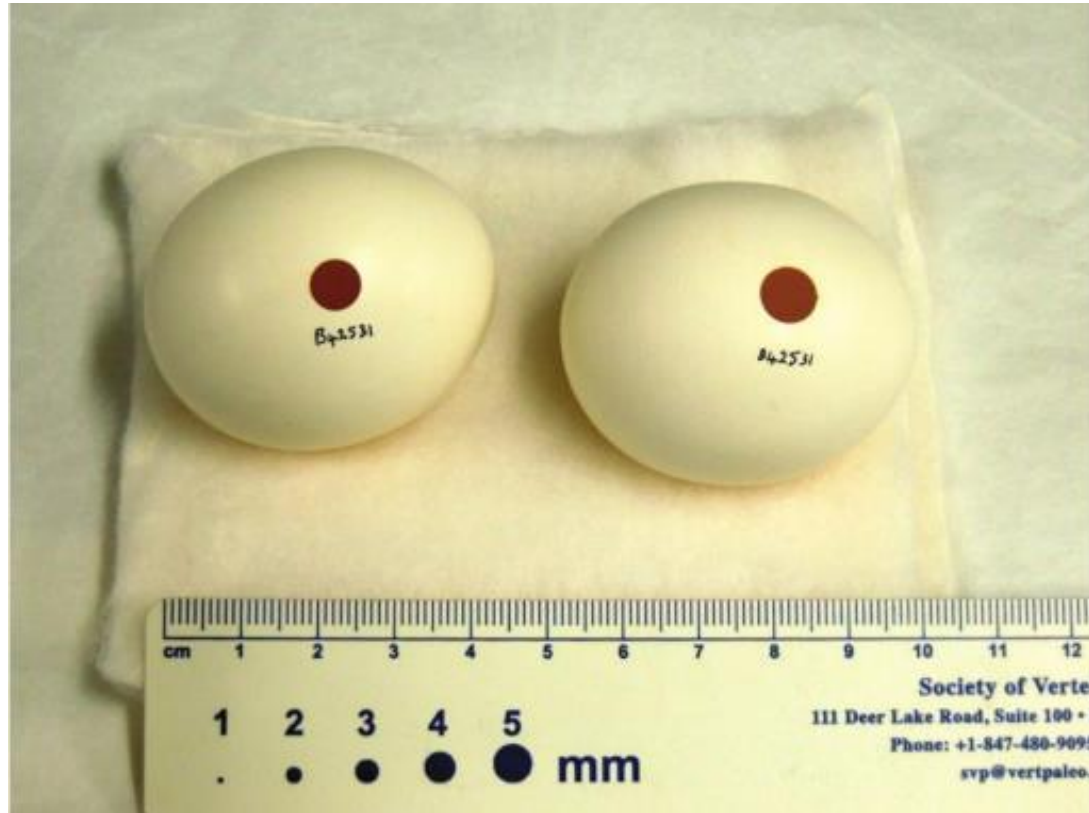
Collecting 2D Data



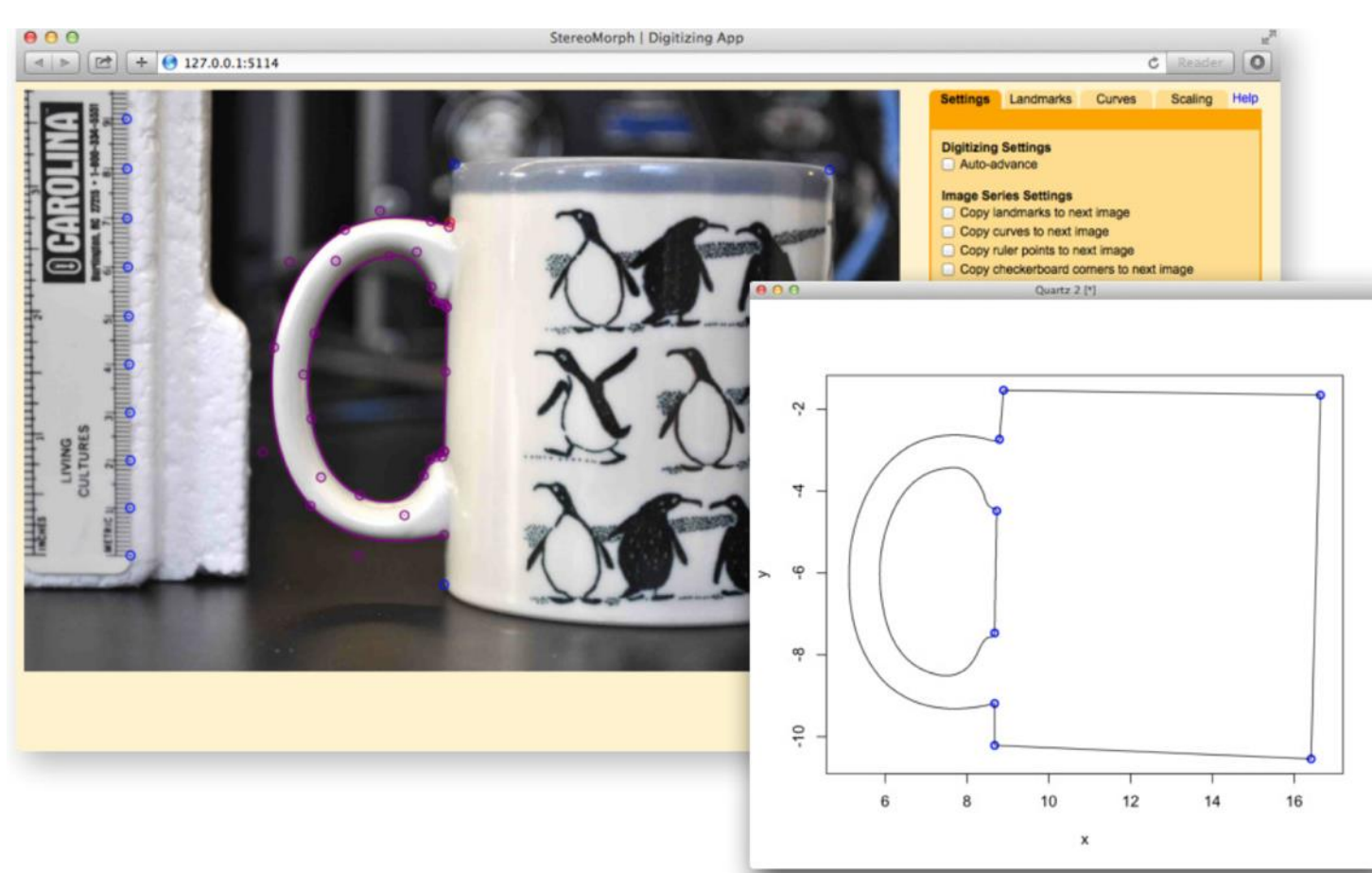
Collecting 2D Data



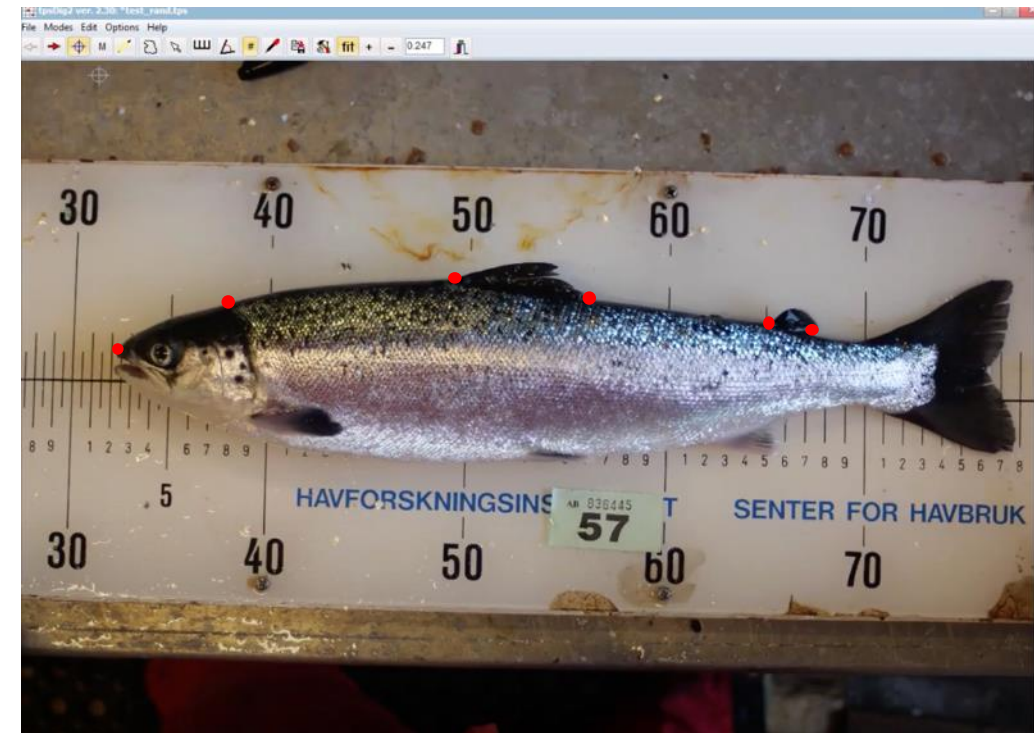
Collecting 2D Data



Collecting 2D Data

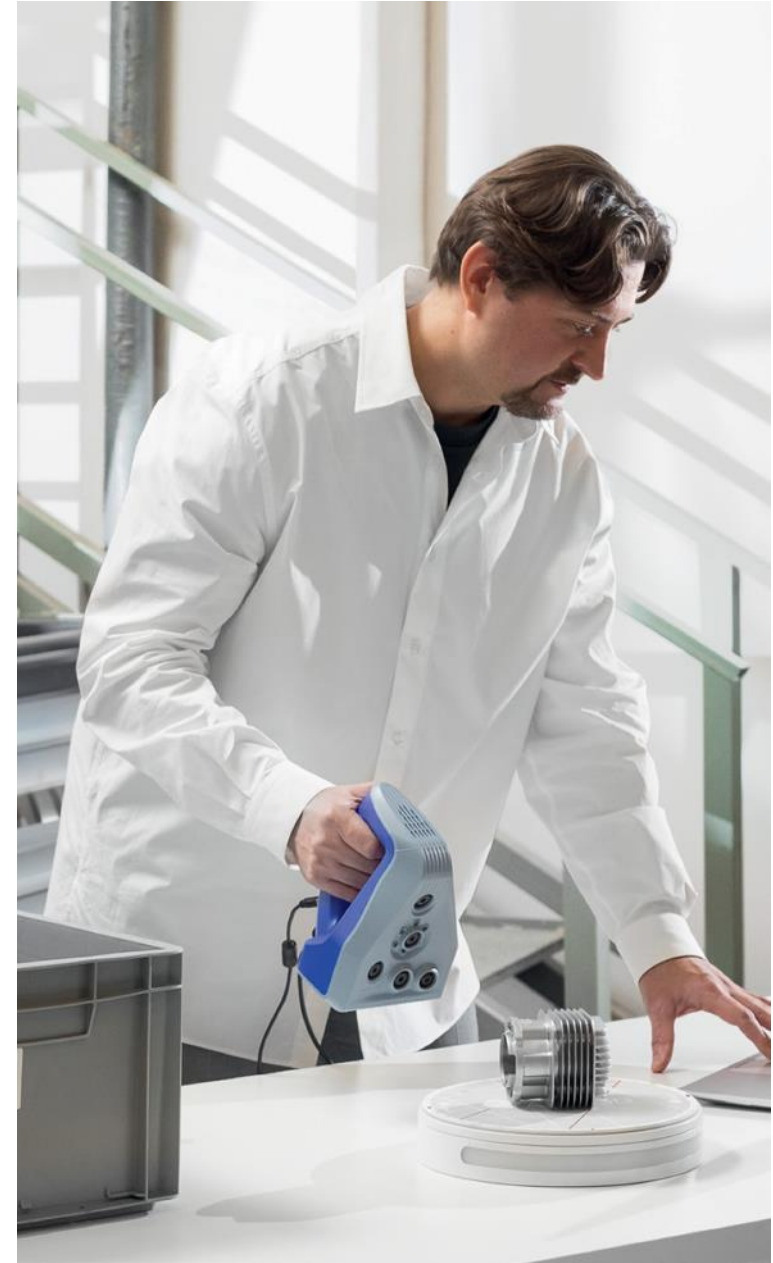


Stereomorph R package

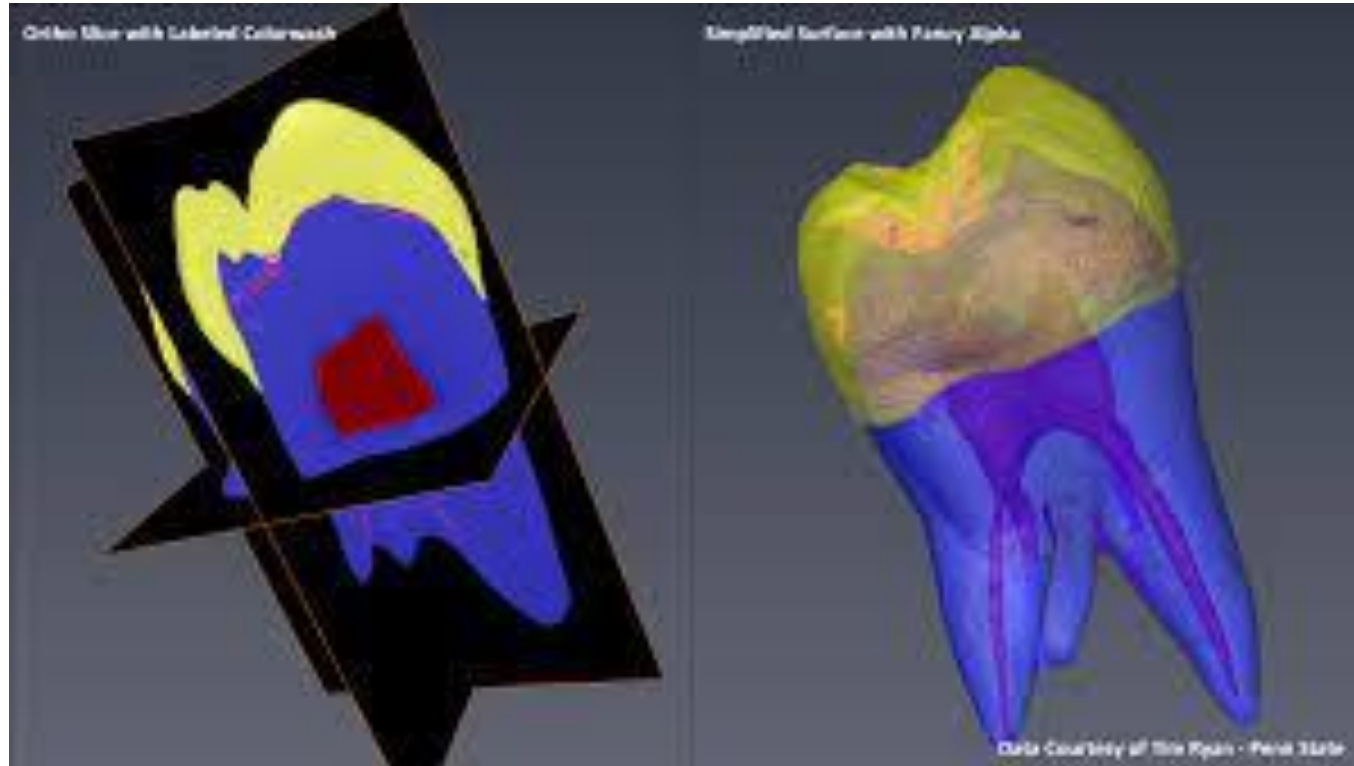


TPS-dig (Windows only)

Collecting 3D Data



Collecting 3D Data

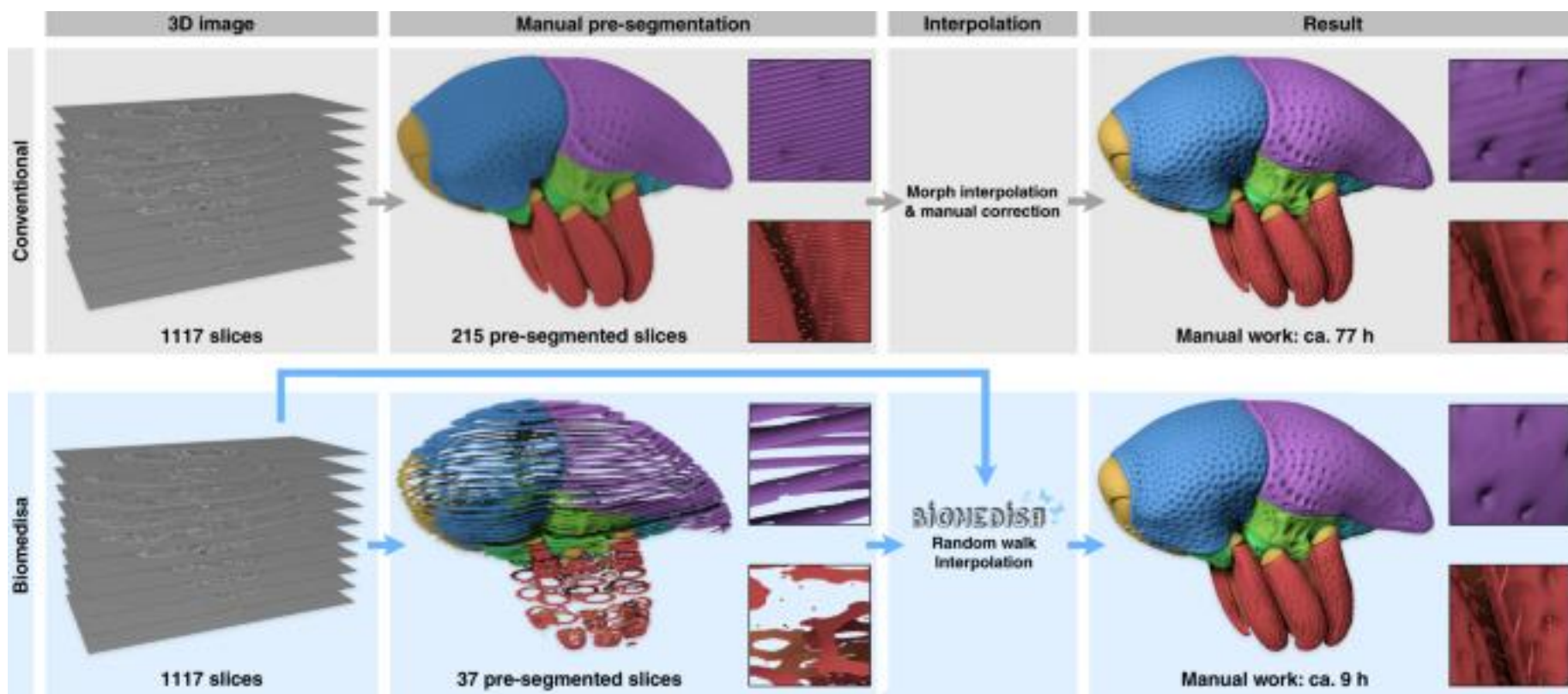


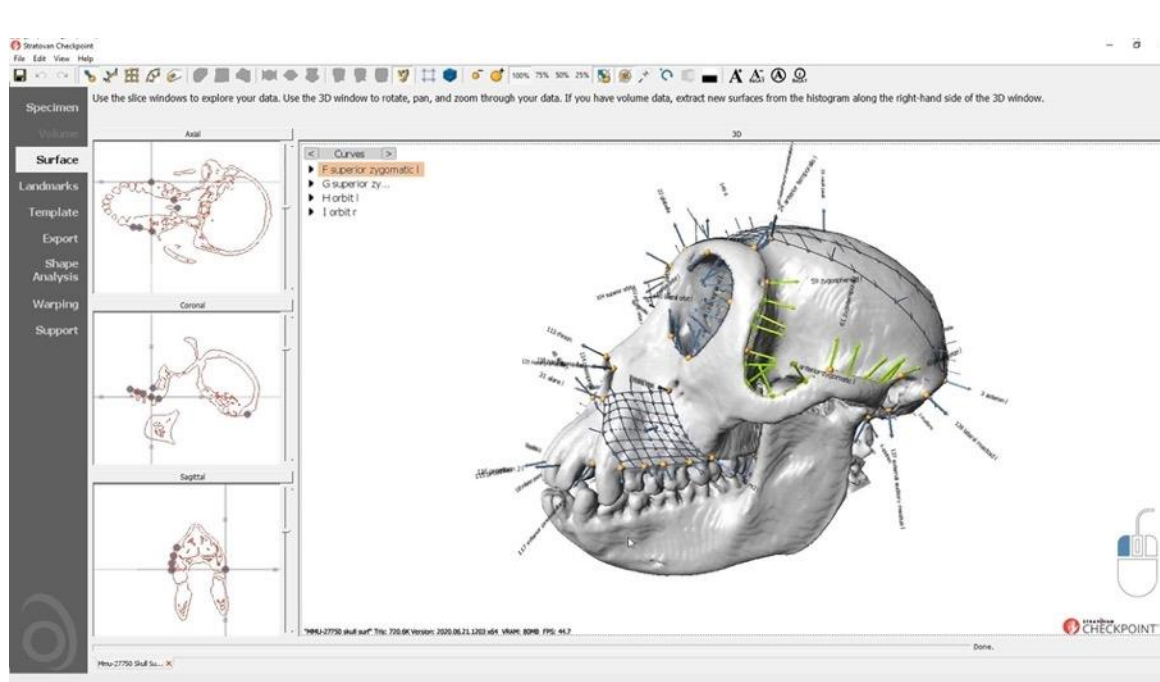
But Why Geometric Morphometrics?

- What can we learn using these method?

Geometric morphometrics today and in the future

- Part 1: AI/Automated segmentation
 - Dragonfly, Biomedisa, SPROUT
- Part 2: landmark-free and automated geometric morphometrics
 - Alpaca, DAA

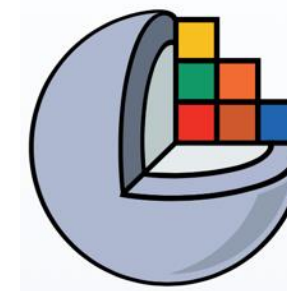
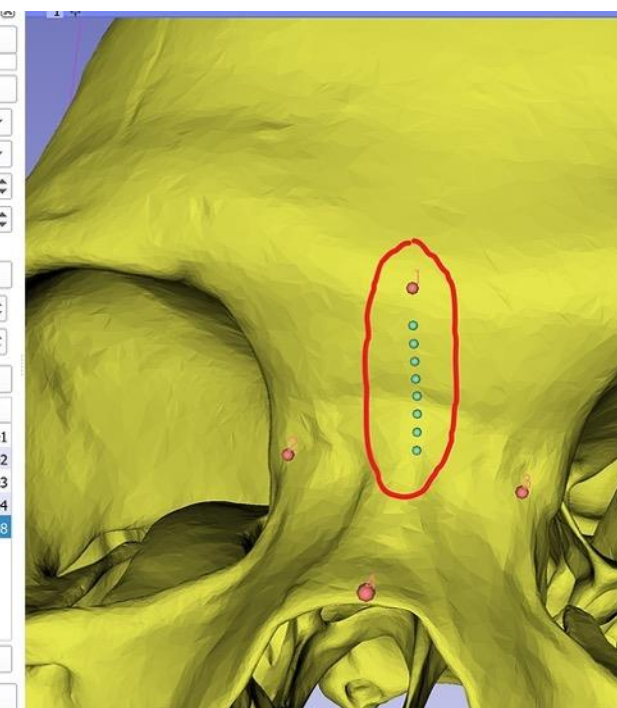
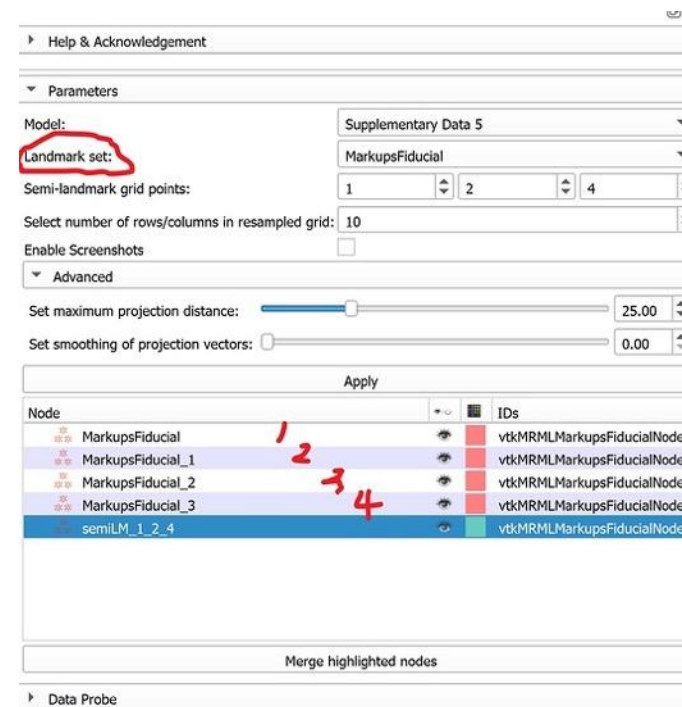




STRATOVAN™



Checkpoint (Windows Only, expensive)



SlicerMorph (multiplatform, free)

Recommended References

